MAT224 - Commutative Algebra

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Intro

These are my own personal notes in commutative algebra, a course I will hopefully be taking in the fall of 2026. Note that as these are for myself there might be gaps, inconsistencies and whatnot which might make it hard to follow so keep that in mind if you are considering using my notes for yourself.

1 Rings

Throughout the course, and these notes, a ring means a commutative ring with unity (meaning multiplicative identity). So every ring R has distinguished elements $1_R, 0_R \in R$ and multiplication satisfies $ab = ba$.

Definition 1.1 ((Commutative Unital) Ring)**.** A ring is a set A together with binary operations $+, \cdot : A \times A \to A$ such that

1. $(A, +)$ is an abelian group with identity 0,
2. multiplication is associative,
3. multiplication is commutative,
4. multiplication distributes over addition,
5. there exists $1 \in A$ such that $1a = a$ for all $a \in A$.

Example 1.1. Examples of familiar rings include

$$\mathbb{Z},\ \mathbb{Q},\ \mathbb{R},\ \mathbb{C}$$

with the usual addition and multiplication.

If A is a ring, then the polynomial ring $A[x]$ is also a ring. More generally, $A[x_1, \dots, x_n]$ is the ring of polynomials in n commuting variables with coefficients in A.

Definition 1.2 (Ring homomorphism)**.** Let A and B be rings. A function

$$\phi : A \to B$$

is a ring homomorphism if

1. $\phi(1_A) = 1_B$,
2. $\phi(a + a') = \phi(a) + \phi(a')$,
3. $\phi(aa') = \phi(a)\phi(a')$.

If ϕ is a bijection, then it is called an isomorphism.

The condition $\phi(1_A) = 1_B$ is a convention that one may or may not choose, but in this course it is not optional, and we say all homomorphisms must explicitly satisfy this.

Proposition 1.1. *Let A be a ring. There exists a unique ring homomorphism*

$$\iota : \mathbb{Z} \to \mathbb{A}.$$

It is given by

$$\iota(n) = n \cdot 1_A.$$

Proof. Since ι is a rinh homomorphism, it must satisfy

$$\iota(1) = 1_A.$$

Then additivity forces

$$\iota(2) = \iota(1+1) = \iota(1) + \iota(1) = 1_A + 1_A,$$

and similarly

$$\iota(n) = \sum_{k=1}^{n} 1_A = \underbrace{1_A + \cdots + 1_A}_{\text{n times}}$$

for $n > 0$. Also,

$$\iota(-n) = -\iota(n).$$

This there is at most one such homomorphism. It is straightforward to check that the function $n \mapsto n \cdot 1_A$ preserves addition, multiplication, and sends 1 to 1_A. Hence it is a ring homomorphism. ∎

Definition 1.3 (Characteristic)**.** The characteristic of a ring A, denoted $\text{Char}(A)$, is the unique integer $n \geq 0$ such that

$$\ker(\mathbb{Z} \to A) = (n).$$

Example 1.2. Here are the characteristics of a few familiar rings:

- $\text{Char}(\mathbb{Z}) = 0$,
- $\text{Char}(\mathbb{F}_p) = p$,
- $\text{Char}(\mathbb{Q}) = 0$.

Intuitively, the characteristic is the smallest $n \in \mathbb{Z}^+$ such that $n \cdot 1_A = 0_A$ with $n := 0$ if no such n exists, and in fact, the characteristic is usually defined as such in a first course in abstract algebra.

Proposition 1.2 (Universal property of $\mathbb{Z}$-polynomials). *Let A be a ring and let $a \in A$. There exists a unique ring homomorphism*

$$\phi_a : \mathbb{Z}[x] \to A$$

such that

$$\phi_a(x) = a.$$

It is given by

$$\phi_a\left(\sum_{k=0}^{n} c_k x^k\right) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Proof. A ring homomorphism $\phi : \mathbb{Z}[x] \to A$ must agree with the unique homomorphism $\mathbb{Z} \to A$ on constant polynomials.

If $\phi(x) = a$, then multiplicativity forces

$$\phi(x^n) = a^n.$$

Additivity then forces

$$\phi(c_0 + c_1 x + \cdots + c_n x^n) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Thus there is at most one such homomorphism.

Conversely, the formula above defines a ring homomorphism, by the usual rules for polynomial addition and multiplication. ■

So homomorphism $\mathbb{Z}[x] \to A$ are the same thing as choices of an element $a \in A$. Intuitively, $\mathbb{Z}[x]$ is the "freest" ring generated by one element.

Definition 1.4 (Ideal). Let A be a ring. An ideal $I \subseteq A$ is a subset such that:

1. I is an additive subgroup of A,
2. for every $a \in A$ and every $x \in I$, we have

$$ax \in I.$$

Definition 1.5 (Principal ideal). Let A be a ring and $a \in A$. The principal ideal by a is

$$(a) = \{ra \mid r \in A\}.$$

An ideal is called principal if it is of the form (a) for some $a \in A$.

More generally, if $a_1, \ldots, a_n \in A$, then

$$(a_1, \ldots, a_n) = \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in A\}.$$

This is the smallest ideal containing all of $a_1, \dots, a_n$.

Example 1.3. $(x, y) \subseteq k[x, y]$ is such an ideal.

$$(x, y) = \{f(x, y)x + g(x, y)y \mid f, g \in k[x, y]\}.$$

It is not the whole ring. It consists precisely of polynomials with zero constant term:

$$(x, y) = \{h \in k[x, y] \mid h(0, 0) = 0\}.$$

Example 1.4. $(0) = \{0\}$ is a (very uninteresting) ideal. $(1) = A$ is another (very uninteresting) ideal. For $n \in \mathbb{Z}$ we have that $(n) = n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Proposition 1.3. *Let $\phi : A \to B$ be a ring homomorphism. Then*

$$\ker \phi = \{a \in A \mid \phi(a) = 0\}$$

is an ideal of A.

Proof. First we show that $\ker \phi$ is an additive subgroup. Since $\phi(0) = 0$, we have $0 \in \ker \phi$. If $x, y \in \ker \phi$, then

$$\phi(x - y) = \phi(x) - \phi(y) = 0 - 0 = 0.$$

Hence $x - y \in \ker \phi$. Now let $a \in A$ and $x \in \ker \phi$. Then

$$\phi(ax) = \phi(a)\phi(x) = \phi(a) \cdot 0 = 0.$$

Thus $ax = \ker \phi$. Therefore $\ker \phi$ is an ideal. ∎

Definition 1.6 (Quotient ring)**.** Let $I \subseteq A$ be an ideal. The quotient ring A/I is the set of additive cosets

$$a + I$$

with addition and multiplication defined by

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Proposition 1.4. *If $I \subseteq A$ is an ideal, then A/I is a ring.*

Proof. The quotient A/I is already an abelian group under addition because I is an additive subgroup.

We need to check multiplication is well-defined.

Suppose

$$a + I = a' + I$$

and

$$b + I = b' + I.$$

Then

$$a - a' \in I, \qquad b - b' \in I.$$

We want to prove

$$ab + I = a'b' + I,$$

i.e.

$$ab - a'b' \in I.$$

Compute:

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b').$$

Since $a - a' \in I$, and I absorbs multiplication by elements of A, we have

$$(a - a')b \in I.$$

Similarly,

$$a'(b - b') \in I.$$

Since I is closed under addition,

$$ab - a'b' \in I.$$

Therefore multiplication is well-defined.

The ring axioms for A/I then follow from the ring axioms in A. ■

The quotient map is

$$\pi : A \to A/I,\ a \mapsto a + I.$$

Its kernel is exactly I:

$$\ker \pi = I.$$

Theorem 1.1 (Fundamental homomorphism theorem)**.** *Let*

$$\phi : A \to B$$

be a ring homomorphism. Then

$$\operatorname{im} \phi$$

is a subring of B, and there is an isomorphism

$$A/\ker \phi \cong \operatorname{im} \phi$$

given by

$$a + \ker \phi \mapsto \phi(a).$$

Proof. Define

$$\overline{\phi} : A/\ker \phi \to \operatorname{im} \phi$$

by

$$\overline{\phi}(a + \ker \phi) = \phi(a).$$

We first check well-definedness.

Suppose

$$a + \ker \varphi = b + \ker \varphi.$$

Then

$$a - b \in \ker \varphi.$$

Thus

$$\varphi(a - b) = 0.$$

So

$$\varphi(a) - \varphi(b) = 0,$$

and hence

$$\varphi(a) = \varphi(b).$$

Therefore $\overline{\varphi}$ is well-defined.

It is a ring homomorphism because φ is.

It is surjective onto $\operatorname{im}\varphi$ by definition.

Finally, its kernel is trivial. Indeed,

$$\overline{\varphi}(a + \ker\varphi) = 0$$

if and only if

$$\varphi(a) = 0,$$

if and only if

$$a \in \ker\varphi,$$

if and only if

$$a + \ker\varphi = 0 + \ker\varphi.$$

Thus $\overline{\varphi}$ is injective. Hence it is an isomorphism. ■

Example 1.5.

$$\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}(i) = \{a + bi \mid a, b \in \mathbb{Z}\}$$

by $f \mapsto f(i)$.
We also have

$$k[x,y]/(x) \cong k(y)$$

by $f \mapsto f(0,y)$.
Lastly

$$k[x,y]/(x,y) \cong k$$

by $f \mapsto f(0,0)$.

Theorem 1.2. *Let $I \subseteq A$ be an ideal. There is a bijection*

$$\{\text{ideals of } A/I\} \leftrightarrow \{\text{ideals } J \subseteq A \text{ such that } I \subseteq J\}.$$

The maps are

$$K \subseteq A/I \mapsto \pi^{-1}(K) \subseteq A,$$

and

$$J \subseteq A, I \subseteq J \mapsto J/I \subseteq A/I.$$

Proof. Let

$$\pi : A \to A/I$$

be the quotient map. First, if $K \subseteq A/I$ is an ideal, then $\pi^{-1}(K)$ is an ideal of A. Since $0 \in K$, and $\ker \pi = I$, we have

$$I \subseteq \pi^{-1}(K).$$

Conversely, suppose $J \subseteq A$ is an ideal with $I \subseteq J$. Then

$$J/I = \{j + I \mid j \in J\}$$

is an ideal of A/I. Now check that these constructions are inverse. If $K \subseteq A/I$, then

$$\pi(\pi^{-1}(K)) = K$$

because π is surjective. If $J \subseteq A$ contains I, then

$$\pi^{-1}(J/I) = J.$$

Indeed, $a \in \pi^{-1}(J/I)$ if and only if

$$a + I \in J/I,$$

which happens if and only if there exists $j \in J$ such that

$$a + I = j + I.$$

This means

$$a - j \in I.$$

Since $I \subseteq J$ and $j \in J$, we get $a \in J$. Hence

$$\pi^{-1}(J/I) = J.$$

Therefore the two maps are inverse bijections. ∎

Example 1.6. The ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to ideals of $\mathbb{Z}$ containing (12). The ideals of $\mathbb{Z}$ are precisely (n) for $n \geq 0$. We need $(12) \subseteq (n)$. This happens precisely when $n \mid 12$. So the ideals of $\mathbb{Z}/12\mathbb{Z}$ are

$$(1), (2), (3), (4), (6), (12) = (0).$$

Proposition 1.5. *Let k be a field. The ideal*

$$(x, y) \subseteq k[x, y]$$

is not principal.

Proof. Suppose for a contradiction that

$$(x, y) = (f)$$

for some $f \in k[x, y]$. Since

$$x \in (x, y) = (f),$$

we have $f \mid x$. Since

$$y \in (x, y) = (f),$$

we have $f \mid y$. However, x and y have no common non-unit factor in $k[x, y]$. Hence f must be a unit. If f is a unit, then

$$(f) = (1) = k[x, y].$$

But this is impossible, because

$$1 \notin (x, y).$$

Therefore (x, y) is not principal. ■

Thus $k[x]$ is a principal ideal domain when k is a field, but $k[x, y]$ is not.

Exercises

Problem 1.1. Prove that if A is a ring with $1 = 0$, then A is the trivial ring.

Proof. Let $a \in A$. Then

$$\begin{aligned} a &= a \\ 1 \cdot a &= 0 \cdot a \\ a &= 0. \end{aligned}$$

Thus

$$A = \{0\}.$$

■

Problem 1.2. Let A be a ring. Prove that there is a unique homomorphism $\mathbb{Z} \to A$.

Proof. Let A be a ring and let $\phi, \psi : \mathbb{Z} \to A$ be homomorphisms.

Then we require

$$\phi(1) = \psi(1) = 1_A.$$

Therefore, for $n > 0$, we have

$$\phi(n) = \phi\left(\sum_{k=1}^{n} 1\right) = \sum_{k=1}^{n} 1_A = \psi\left(\sum_{k=1}^{n} 1\right) = \psi(n).$$

For $n < 0$ we have $n = -m$ for some $m > 0$. Then

$$\phi(n) = \phi(-m) = -\phi(m) = -\sum_{k=1}^{n} 1_A = -\psi(m) = \psi(-m) = \psi(n).$$

Lastly,

$$\phi(0) = 0 = \psi(0).$$

Thus ϕ and ψ agree on all of $\mathbb{Z}$ and hence

$$\psi = \phi.$$

■

Problem 1.3. Identify $k[x, y]/(x, y - 1)$.

Solution. Let $\phi : k[x, y] \to k$ by

$$\phi(f) = f(0, 1).$$

Then ϕ is a ring homomorphism and furthermore it is surjective since, for $a \in k$ we have $f(x, y) := a \mapsto a$.

Now we show that $\ker \phi = (x, y - 1)$. First of all $x, y - 1 \in \ker \phi$ since $x(0, 1) = 0$ and $y - 1(0, 1) = 0$. Thus

$$(x, y - 1) \subseteq \ker \phi.$$

Now suppose $f \in \ker \phi$. Then $f(0, 1) = 0$ so $f(x, y) = xg(x, y) + (y - 1)h(x, y) + f(0, 1)$. Since $f(0, 1) = 0$ we have $f(x, y) = xg(x, y) + (y - 1)h(x, y)$ so $f(x, y) \in (x, y - 1)$. Thus $\ker \phi \subseteq (x, y - 1)$. The two together give $(x, y - 1) = \ker \phi$. Thus, by the fundamental theorem of ring homomorphisms (or first isomorphism theorem, if you prefer), we have

$$\frac{k[x, y]}{(x, y - 1)} \cong k.$$

■

Problem 1.4. List all ideals of $\mathbb{Z}/18\mathbb{Z}$.

Solution. The ideals are

- (0),
- (1),
- (2),
- (3),
- (6),
- (9),

i.e. (n) where $n \mid 18$. Note that $(0) = (18)$ and 18 divides 18. ■

2 Special ideals and rings

Definition 2.1. Let A be a ring.
An element $u \in A$ is a **unit** if there exists $v \in A$ such that

$$uv = 1.$$

An element $x \in A$ is a **zero-divisor** if there exists a nonzero element $y \in A$ such that

$$xy = 0.$$

A ring A is an **integral domain** if

$$A \neq 0$$

and 0 is the only zero-divisor.
Equivalently, A is an integral domain if

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0.$$

A ring A is a **field** if

$$A \neq 0$$

and every nonzero element of A is a unit.

Example 2.1. $\mathbb{Z}$ is an integral domain, but not a field. The rings

$$\mathbb{Q},\ \mathbb{R},\ \mathbb{C}$$

are fields. The ring $\mathbb{Z}/4\mathbb{Z}$ is not an integral domain.

Proposition 2.1. *Let A be a ring and $x \in A$. Then*

$$x \text{ is a unit}$$

if and only if

$$(x) = (1).$$

Proof. Suppose first x is a unit. Then $\exists y \in A$ s.t.

$$xy = 1.$$

Therefore $1 \in (x)$. Since every ideal containing 1 is the whole ring, we get

$$(x) = A = (1).$$

Conversely, suppose $(x) = (1)$. Then

$$1 \in (x).$$

By definition of the principal ideal (x), there exists $y \in A$ such that

$$1 = yx.$$

Since A is commutative,

$$xy = 1.$$

Therefore x is a unit. ■

Theorem 2.1. *Let A be a ring. Then A is a field if and only if A has exactly two ideals:*

$$(0) \neq (1).$$

Proof. Suppose A is a field.

Let $I \subseteq A$ be a nonzero ideal. Choose

$$0 \neq x \in I.$$

Since A is a field, x is a unit. Then

$$(x) = (1).$$

But

$$(x) \subseteq I.$$

Therefore

$$(1) = A \subseteq I$$

so

$$I = (1).$$

So the only ideals are

$$(0),\ (1).$$

Conversely, suppose A has two ideals

$$(0) \neq (1).$$

Let $0 \neq x \in A$. Then

$$(x) \neq (0).$$

Since the only nonzero ideal is (1), we have

$$(x) = (1).$$

Then x is a unit and thus since every nonzero element of A is a unit, A is a field. ■

Definition 2.2 (Prime ideal)**.** Let A be a ring and let $I \subseteq A$ be an ideal. We say I is a **prime ideal** if

$$I \neq (1)$$

and whenever

$$ab \in I,$$

we have

$$a \in I \quad \text{or} \quad b \in I.$$

Equivalently,

$$a, b \notin I \implies ab \notin I.$$

The set of all prime ideals of A is denoted

$$\operatorname{Spec}(A).$$

Example 2.2. The prime ideals of $\mathbb{Z}$ are (0) and (p) where p is a prime number.

Definition 2.3 (Maximal ideal)**.** Let A be a ring and let $I \subseteq A$ be an ideal. We say I is a **maximal ideal** if

$$I \neq (1)$$

and the only ideals J satisfying

$$I \subseteq J \subseteq A$$

are

$$J = I$$

and

$$J = A.$$

Equivalently, there is no ideal strictly between I and A.
The set of maximal ideals of A is denoted

$$\operatorname{Specm}(A).$$

Example 2.3. The maximal ideals of $\mathbb{Z}$ are (p).

Theorem 2.2. *Let A be a ring and let $I \subseteq A$ be an ideal. Then*

$$I \text{ is prime}$$

if and only if

$$A/I \text{ is an integral domain.}$$

Also,

$$I \text{ is maximal}$$

if and only if

$$A/I \text{ is a field.}$$

Exercises

Problem 2.1. Decide whether $(2) \subseteq \mathbb{Z}/6\mathbb{Z}$ is prime, maximal, both, or neither.

Solution. It is maximal. Let $\phi : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ by

$$\phi(a) = a \mod 2.$$

Then this map is surjective and $\ker \phi = (2)$. By the first isomorphism theorem we have

$$\mathbb{Z}_6/\ker \phi \cong \mathbb{Z}_2$$

which is a field and thus $\ker \phi$ is maximal. ■

Problem 2.2. Decide whether $(x-1) \subseteq k[x]$ is prime or maximal.

Solution. Let $\phi : k[x] \to k$ by

$$\phi(f) = f(1).$$

Then $\ker \phi = (x-1)$, the map is surjective so it follows as from before that $(x-1)$ is maximal. ■

3 Krull's theorem, PIDs/UFDs, local rings

Theorem 3.1 (Krull's theorem)**.** *Let A be a nonzero ring. Then A has a maximal ideal.*
Equivalently:

$$A \neq 0 \Rightarrow \operatorname{Specm}(A) \neq \emptyset.$$

This is not at all obvious for arbitrary rings, because rings may have infinitely many ideals and no obvious "largest" proper ideal. To prove Krull's theorem we will need the following set-theoretic principle.

Lemma 3.1 (Zorn's lemma)**.** *Let S be a partially ordered set. Suppose every chain in S has an upper bound in S. Then S has a maximal element.*

Here:

- a **chain** is a subset where any two elements are comparable,
- an **upper bound** for a chian $C \subseteq S$ is an element $u \in S$ such that $c \leq u$ for all $u \in C$,
- a **maximal element** is an element $m \in S$ such that there is no $s \in S$ with $m < s$.

Proof of Krull's theorem. Let $A \neq 0$. Consider the set

$$S = \{I \subseteq A \mid I \text{ is an ideal and } I \neq A\}.$$

So S is the set of all proper ideals of A. We partially order S by inclusion:

$$I \leq J \Leftrightarrow I \subseteq J.$$

We want to apply Zorn's lemma.

Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be a chain in S. Define

$$I = \bigcup_{\lambda \in \Lambda} I_\lambda.$$

We claim $I \in S$, meaning I is a proper ideal.

First, I is an ideal.

Let $a, b \in I$. Then there exists $\lambda, \mu \in \Lambda$ such that

$$a \in I_\lambda,\ b \in I_\mu.$$

Since I_μ is an ideal,

$$a - b \in I_\mu \subseteq I.$$

Thus I is an additive subgroup. Now let $r \in A$ and $a \in I$. Then $a \in I_\lambda$ for some λ, so

$$ra \in I_\lambda \subseteq I.$$

Thus I absorbs multiplication from A and is therefore an ideal. Second $I \neq A$. If $I = A$, then in particular

$$1 \in I.$$

So $1 \in I_\lambda$ for some λ, but then

$$I_\lambda = A,$$

contradicting $I_\lambda \in S$. Hence $I \neq A$.

So $I \in S$, and I is an upper bound for the chain. Therefore every chain in S has an upper bound. By Zorn's lemma, S has a maximal element m. This m is a proper ideal of A, and no proper ideal strictly contains it. Hence m is a maximal ideal.

Thus every nonzero ring has a maximal ideal. ■

Corollary 3.1. *Let $I \subseteq A$ be a proper ideal. Then there exists a maximal ideal $m \subseteq A$ such that*

$$I \subseteq m.$$

Proof. Since $I \neq A$, the quotient

$$A/I$$

is nonzero. By Krull's theorem, A/I has a maximal ideal $\mathfrak{m}$. Let

$$\pi : A \to A/I$$

be the quotient map. Then

$$m = \pi^{-1}(\mathfrak{m})$$

is a maximal ideal of A containing I, by the correspondence theorem. ■

Corollary 3.2. *Let A be a ring and $x \in A$. Then*

$$x \text{ is a unit}$$

if and only if

$$x \notin m$$

for every maximal ideal $m \subseteq A$.
Equivalently:

$$x \text{ is not a unit} \Leftrightarrow x \in m \text{ for some maximal ideal } m.$$

Proof. We have:

$$x \text{ is not a unit} \Leftrightarrow (x) \neq A.$$

By the previous Corollary,

$$(x) \neq A \Leftrightarrow (x) \subseteq m$$

for some maximal ideal m. But

$$(x) \subseteq m \Leftrightarrow x \in m.$$

Therefore

$$x \text{ is not a unit} \Leftrightarrow x \in m$$

for some maximal ideal m. ■

Definition 3.1 (PID)**.** An integral domain A is a **principal ideal domain**, or PID, if every ideal of A is principal.

Example 3.1. Some examples of PIDs include

- $k[x]$ for any field k,
- any field k,
- $\mathbb{Z}$.

Definition 3.2. Let A be an integral domain. Let $f \in A$ be neither 0 nor a unit. We say f is **irreducible** if whenever

$$f = gh,$$

then either g is a unit or h is a unit, meaning f cannot be factored nontrivially.

Definition 3.3 (UFD)**.** An integral domain A is a **unique factorization domain**, or UFD, if every nonzero nonunit $f \in A$ can be written as a finite product of irreducibles,

$$f = p_1 \cdots p_n,$$

and this factorization is unique up to reordering and multiplication by units.

Example 3.2. Some examples include:

- $\mathbb{Z}$,

- $k[x]$ for any field k,
- $k[x_1, \dots, x_n]$ for any field k.

We remark that PID $\Rightarrow$ UFD and that if A is a UFD then $A[x]$ is a UFD.

Definition 3.4 (Local ring)**.** A ring A is **local** if it has exactly one maximal ideal. That maximal ideal is usually denoted

$$\mathfrak{m}.$$

So a local ring is a pair $(A, \mathfrak{m})$, where $\mathfrak{m}$ is the unique maximal ideal.

Example 3.3. Every field k is local since the only maximal ideal is (0).
$\mathbb{Z}$ is a nonexample as it has maximal ideals (p) for every prime p.

Proposition 3.1. *Let A be a local ring with unique maximal ideal $\mathfrak{m}$. Then the set of units in A is*

$$A \setminus \mathfrak{m}.$$

Equivalently:

$$x \in \mathfrak{m} \Leftrightarrow x \text{ is not a unit.}$$

Proof. By the corollary to Krull's theorem, an element $x \in A$ is a unit if and only if it is contained in no maximal ideal. But A has exactly one maximal ideal, namely $\mathfrak{m}$. Therefore

$$x \text{ is a unit} \Leftrightarrow x \notin \mathfrak{m}.$$

So the units are exactly

$$A \setminus \mathfrak{m}.$$

■

Proposition 3.2. *Let A be a ring. Suppose the set of nonunits*

$$\mathfrak{m} = \{x \in A \mid x \text{ is not unit}\}$$

is an ideal. Then A is local, and $\mathfrak{m}$ is its unique maximal ideal.

Proof. Let $I \subseteq A$ be a proper ideal. We claim

$$I \subseteq \mathfrak{m}.$$

Indeed, if $x \in I$ were a unit, then

$$1 \in I,$$

because

$$x^{-1}x = 1.$$

Then $I = A$, contradicting that I is proper. So every element of I is a nonunit, hence

$$I \subseteq \mathfrak{m}.$$

Thus every proper ideal of A is contained in $\mathfrak{m}$. Since $\mathfrak{m}$ is itself a proper ideal, it is the unique maximal ideal. Therefore A is local. ■

Exercises

Problem 3.1.

(a) Let A be a ring, and let $f = a_n x^n + \cdots + a_0 \in A[x]$. Prove that if f is a unit of $A[x]$, then a_0 is a unit of A.

(b) Show that if a is a unit of A, then the constant polynomial $a \in A[x]$ is a unit of $A[x]$.

(c) Show that if A is an integral domain, then the units of $A[x]$ are exactly the constant polynomials a with a a unit of A.

Proof of (a). Suppose

$$f = \sum_{k=0}^{n} a_k x^k$$

is a unit of $A[x]$. Then there exists

$$g = \sum_{k=0}^{m} b_k x^k$$

such that

$$fg = 1.$$

Let $\phi : A[x] \to A$ by

$$\phi(f) = f(0).$$

Then ϕ is a ring homomorphism, and thus

$$1_A = \phi(1_{A[x]}) = \phi(fg) = a_0 b_0$$

so a_0 is a unit in A. ■

Proof of (b). Suppose $a \in A$ is a unit. Then there exists $b \in A$ so that

$$ab = 1_A.$$

Then $a, b \in A[x]$ as constant polynomials. Once again

$$1_{A[x]} = 1_A = ab = \phi(a)\phi(b) = \phi(ab).$$

■

Proof. From the above we have that units in A as constant polynomials are units in $A[x]$.

Now let $f \in A[x]$ be a unit. Then there is $g \in A[x]$ so

$$fg = 1.$$

Then since $\deg(g \cdot h) = \deg(g) + \deg(h)$ we have

$$\deg(f) + \deg(g) = \deg(f \cdot g) = \deg(1) = 0.$$

Then $\deg(f) = \deg(g) = 0$ so they are constant polynomials.

We require A to be an integral domain because otherwise we could take $A = \mathbb{Z}_4$ and $f = 2x + 1$. Then $f^2 = 1$. ■

Problem 3.2. Prove that a ring A is a field if and only if the following holds: $A \neq 0$ and every homomorphism from A to a non-zero ring is injective.

Proof. Let A and $B \neq 0$ be rings.

First assume A is a field. Then $\ker \phi$ is A or (0) for every homomorphism $\phi : A \to B$. If $\ker \phi = A$ then $\psi(x) = 0_B$ for all $x \in A$ and in particular $1_B = \phi(1_A) = 0_B$ so B is the trivial ring, but this contradicts our assumption. Thus $\ker \phi = (0)$, meaning ϕ is injective. Furtheremore, by definition, $A \neq 0$.

Now assume $A \neq 0$ and every homomorphism $\phi : A \to B$ has $\ker \phi = \{0_A\}$ when B is nonzero.

Let $I \subseteq A$ is some ideal. Consider the canonical quotient homomorphism

$$\pi : A \to A/I \text{ defined by } \pi(a) = a + I.$$

Now either A/I is nonzero or not. If it is the zero ring then $\ker \pi = A$. Otherwise if A/I is nonzero, then by assumption $\ker \pi = (0)$. By construction,

$$\ker \pi = I$$

so every ideal I of A is either (0) or A. Hence A is a field. ■

Problem 3.3. Let A be a ring, and let $x \in A$. Consider the map $m_x : A \to A$ given by $m_x(y) = xy$ for some $x \in A$.

(a) Show that m_x is a ring homomorphism if and only if $x = 1$.

(b) Show that x is a 0-divisor if and only if m_x is non-injective.
(c) Show that x is a unit if and only if m_x is surjective.

Proof of (a). Suppose $x = 1$. Then $m_x : A \to A$ is the identity map $\mathrm{id} : A \to A$,

$$\mathrm{id}(y) = y,$$

which is a homomorphism.

Now suppose m_x is a ring homomorphism and suppose for a contradiction that $x \neq 1$. Then

$$m_x(1) = x \neq 1$$

so m_x does not preserve the multiplicative identity and thus fails to be a homomorphism. ■

Proof of (b). Suppose x is a 0-divisor, meaning $x \neq 0$ and there exists $\xi \in A \setminus \{0\}$ so that

$$x \cdot \xi = 0.$$

Then $\xi \in \ker m_x$ and $\xi \neq 0$ so m_x cannot be injective.

Suppose m_x is not injective. So $|\ker m_x| > 1$. Then there exists $\xi \in \ker m_x \setminus \{0\}$. Then

$$m_x(\xi) = m_x(0)$$
$$x \cdot \xi = x \cdot 0 = 0$$

so x is a 0-divisor. ■

Proof of (c). Suppose x is a unit. Then there exists $x^{-1} \in A$ so that $xx^{-1} = 1$. Then, for any $a \in A$ we see that

$$xx^{-1}a \overset{m_x}{\mapsto} a$$

so m_x is surjective.

Now suppose m_x is surjective. Then, for any $a \in A$ there exists $b \in A$ so that

$$m_x(b) = xb = a.$$

In particular this holds for $1 \in A$ so there exists $x^{-1} \in A$ so that

$$m_x(x^{-1}) = xx^{-1} = 1.$$

■

Problem 3.4. Let A be a ring, and let $\{P_i\}_{i \in I}$ be a non-empty chain of prime ideals (where "chain" ,eans that for every two $i, j \in I$ either $P_i \subseteq P_j$ or $P_j \subseteq P_i$). Prove that

$\bigcup_{i\in I} P_i$ and $\bigcap_{i\in I} P_i$ are prime ideals.

Proof. Let

$$J = \bigcup_{i\in I} P_i$$

and

$$K = \bigcap_{i\in I} P_i.$$

We begin with J. Since the chain is nonempty, pick any P_i. Since P_i is an ideal, $0 \in P_i$ so $0 \in J \neq \emptyset$. Let $x \in J$ and $r \in A$. Since $x \in J$, $x \in P_i$ for some specific index i. Since P_i is an ideal we have $rx \in P_i$ so $rx \in J$. Let $x, y \in J$. Then $x \in P_i$ and $y \in P_j$ for some specific indicies $i, j \in I$. The chain-condition gives us that $P_i \subseteq P_j$ or vice-versa. Assume wlog the former holds. Then $x, y \in P_j$ so $x+y \in P_j \subseteq J$. Thus J is closed under addition.

Now since every P_i is prime we have $1 \notin P_i$ for every $i \in I$ so $1 \notin J$. Thus $J \neq A$ and is therefore a proper ideal. Let $a, b \in A$ such that $ab \in J$. Then there is some P_i so that $ab \in P_i$. Since P_i is prime we have $a \in P_i$ or $b \in P_i$.

Now we do K. Similar as before $0 \in P_i$ for all $i \in I$ so $0 \in K \neq \emptyset$. Let $x \in J$ and $r \in A$. Since $x \in J$ we have that $x \in P_i$ for every i so $ax \in P_i$ for every i hence $ax \in K$. Let $x, y \in K$. Then $x, y \in P_i$ for every i hence $x + y \in P_i$ for every i so $x + y \in K$. Thus K is closed under addition.

Now supose $ab \in K$. Then $ab \in P_i$ for every i so $a \in P_i$ or $b \in P_i$ for every i. Thus $a \in K$ or $b \in K$. ■

Problem 3.5. Let X be a topological space and $x \in X$. We consider pairs (U, f), where $U \subseteq X$ is an open set with $x \in U$ and $f : U \to \mathbb{R}$ is a continuous function. Two pairs (U, f) and (V, g) are called equivalent if there exists an open neighborhood W of x with $W \subseteq U \cap V$ such that $f\big|_W = g\big|_W$. An equivalence class is called a germ of a function at x. We denote by $\mathcal{O}_{X,x}$ the set of all such germs.

(a) Show that addition and multiplication of germs are well defined by

$$(U, f) + (V, g) := (U \cap V, f\big|_{U\cap V} + g\big|_{U\cap V}),$$

$$(U, f) + (V, g) := (U \cap V, f\big|_{U\cap V}\ g\big|_{U\cap V}).$$

Verify that $\mathcal{O}_{X,x}$ is a ring with these operations, and describe the elements $0, 1 \in \mathcal{O}_{X,x}$ as germs.

(b) Show that a germ (U, f) is a unit in $\mathcal{O}(X, x)$ if and only if $f(x) \neq 0$.

(c) Show that $\mathcal{O}_{X,x}$ is a local ring, with maximal ideal

$$\mathfrak{m} = \{(U, f) \in \mathcal{O}_{X,x} \mid f(x) = 0\},$$

and residue field

$$\mathcal{O}_{X,x}/\mathfrak{m} \cong \mathbb{R}.$$

Proof of (a). Suppose $(U_1, f_1) = (U_2, f_2)$ and $(V_1, g_1) = (V_2, g_2)$. By definition of the equivalence relation

- there exists an open neighborhood $W_1 \subseteq U_1 \cap U_2$ of x such that $f_1 \big|_{W_1} = f_2 \big|_{W_1}$, and
- there exists an open neighborhood $W_2 \subseteq V_1 \cap V_2$ of x such that $g_1 \big|_{W_2} = g_2 \big|_{W_2}$.

Define $W = W_1 \cap W_2$. Since open neighborhoods are closed under finite intersections we have that W is an open neighborhood of x. Furthermore, $W \subseteq (U_1 \cap v_1) \cap (U_2 \cap V_2)$. Restricting the sum to W we get

$$(f_1 + g_2) \big|_W = f_1 \big|_W + g_1 \big|_W + f_2 \big|_W + g_2 \big|_W = (f_2 + g_2) \big|_W .$$

Thus addition is well-defined and this approach works for multiplication as well.

The ring properties of associativity, commutativity and distributivity are inherited from the pointwise operations of $\mathbb{R}$.

The additive itentity is $(X, \mathbf{0})$ where $\mathbf{0} : X \to \mathbb{R}$ is the constant function 0.

The multiplicative identity is $(X, \mathbf{1})$ where $\mathbf{1} : X \to \mathbb{R}$ is the constant 1. ■

Proof of (b). Suppose (U, f) is a unit. Then there exists (V, g) such that $(U, f) \cdot (V, g) = (X, \mathbf{1})$. Then there is an open neighborhood $W \subseteq U \cap V$ of x where $f(k)g(k) = 1$ for all $k \in W$. Evaluated at x we get

$$f(x)g(x) = 1.$$

Since $f(x), g(x) \in \mathscr{R}$ we cannot have $f(x) = 0$ as 0^{-1} does not exist.

Suppose $f(x) \neq 0$. Since $f : U \to \mathbb{R}$ is continuous and $\{0\}^c = \mathbb{R} \setminus \{0\}$ is an open subset of $\mathbb{R}$, the preimage $V = f^{-1}(\{0\}^c)$ is open in U and thus open in X.

Since $f(x) \neq 0$ we have $x \in V$ so we can define a continuous $g : V \to \mathbb{R}$ by

$$g(k) = \frac{1}{f(k)} \text{ for all } k \in V.$$

Then, on the open neighborhood V, we have $f \big|_V \cdot g = 1 \big|_V$. Thus $[U, f]$ is a unit. ■

4 A step back

Everything thus far has been loosely covering Jørgen V. Rennemo's notes as I was planing to take this course at UiO as I usually do. Now I will be taking the equivalent MAT224 course at UiB instead! Thus I think it is fitting to go back, as there are some pretty big differences in the flavour of this course when compared with MAT4200 of UiO.

Gathmann motivates commutative algebra by telling us that commutative rings provide a common algebraic language for things that initially look number-theoretic. In particular, polynomial equations define geometric objects, and the functions on those objects form commutative rings.

The philosophy is that instead of studying geometric objects directly, study the ring of polynomial functions on it. This is quite a bit more algebraic geometry esque. Very cool!

Definition 4.1. Let K be a field. The affine n-space over K is the set

$$\mathbb{A}^n_K = \{(a_1, \dots, a_n) \mid a_i \in K\}.$$

As a set,

$$\mathbb{A}^n_K = K^n.$$

The different notation emphasizes that we are not interested in the set as a vector space. We are thinking of it simply as a set of points on which polynomial equations can be imposed.

For instance $\mathbb{A}^2_{\mathbb{R}}$ is simply the real plane.

Let $f \in K[x_1, \dots, x_n]$. Write

$$f = \sum_{(i_1,\dots,i_n)\in\mathbb{N}^n} a_{i_1\dots,i_n} x_1^{i_1} \cdots x_n^{i_n},$$

where only finitely many coefficients are nonzero.

Given $a = (a_1, \dots, a_n) \in \mathbb{A}^n_K$, we evaluate f at a by

$$f(a) = \sum a_{i_1,\dots,i_n} a_1^{i_1} \cdots a_n^{i_n}.$$

The distinction is simply between the formal variables x_j and a point with coordinates a_j at which we substitute values.

Definition 4.2. Let

$$S \subseteq K[x_1, \dots, x_n]$$

be any set of polynomials. Its zero locus is

$$V(S) = \{a \in \mathbb{A}^n_K \mid f(a) = 0 \text{ for every } f \in S\}.$$

A subset of $\mathbb{A}^n_K$ that can be written as $V(S)$ is called an **affine variety** using Gathmann's terminology.

If $S = (f_1, \dots, f_r)$ we normally write $V(f_1, \dots, f_r)$.

Example 4.1. Over $\mathbb{R}$,

$$V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$$

is the unit circle. Indeed,

$$(a, b) \in V(x^2 + y^2 - 1)$$

if and only if

$$a^2 + b^2 = 1.$$

Example 4.2. Consider

$$V(xy) \subseteq \mathbb{A}^2_{\mathbb{K}}.$$

A point (a, b) lies in $V(xy)$ iff $ab = 0$. Since K is a field, this means $a = 0$ or $b = 0$. Hence

$$V(xy) = V(x) \cup V(y).$$

Geometrically this is the union of the x and y axes. This notion will appear later algebraically through prime ideals and primary decomposition. How exciting!

Now, why can't we simply go through equations themselves? Suppose $X = V(f)$. It would be tempting to say that f is the equations of X. However, there is an immediate problem:

$$V(f) = V(f^2).$$

Many entirely different collectioons of polynomials can have exactly the same zero locus. Gathmann uses this ambiguity to motivate replacing a chosen set of equations by all equations that vanish on the object.

Definition 4.3. Let $X \subseteq \mathbb{A}^n_K$. We do not actuaklly need to assume X is a variety yet.
The ideal of X is

$$I(X) = \{f \in K[x_1, \dots, x_n] \,:\, f(a) = 0 \text{ for every } a \in X\}.$$

Proposition 4.1.

$$I(x) \triangleleft K[x_1, \dots, x_n].$$

Proof. Ceirtainly $0 \in I(X)$.

Let $f, g \in I(X)$. For every $a \in X$,

$$(f-g)(a) = f(a) - g(a) = 0 - 0 = 0.$$

Hence $f - g \in I(X)$. Thus $I(X)$ is an additive subgroup. Now let $h \in K[x_1, \dots, x_n]$ and $f \in I(X)$. For every $a \in X$,

$$(hf)(a) = h(a)f(a) = h(a) \cdot 0 = 0.$$

Thus $hf \in I(X)$. Therefore $I(X)$ is an ideal. ■

We now take the quotient

$$A(X) = K[x_1, \dots, x_n]/I(X).$$

This is called the **coordinate ring** of X, or the ring of polynomials functions on X.

Why quotient by $I(X)$?

Suppose $f, g \in K[x_1, \dots, x_n]$. In the quotient, $f+I(X) = g+I(X)$ if and only if $f-g \in I(X)$ so $f(a) = g(a)$ for every $a \in X$.

Thus $f = g$ in $A(X)$ if and only if f and g define the same function on X.

Example 4.3. Let $X = V(y - x^2) \subseteq \mathbb{A}^n_K$.
On X we have $y = x^2$. Thus as polynomial functions on X, y and x^2 are the same function.
Algebraically this is encoded by $A(X) = K[x, y]/I(X)$.

There is some nuance here when working over finite fields, but Gathmann says the course will largely surpress this as we will be working over infinite fields like $\mathbb{R}$ and $\mathbb{C}$ for the most part.

We will now introduce the ideal of a point.

Take $a = (a_1, \dots, a_n)$ in the affine n-space over some field K. Then

$$I(a) = I(\{a\}) = (x_1 - a_1, \dots, x_n - a_n).$$

Proposition 4.2. *For a as above, we have*

$$I(a) = (x_1 - a_1, \dots, x_n - a_n).$$

Proof. First suppose $f \in (x_1 - a_1, \dots, x_n - a_n)$.

Then there exist polynomials $g_1, \dots, g_n$ such that

$$f = \sum_{i=1}^{n} (x_i - a_i) g_i.$$

Evaluating at a,

$$f(a) = \sum_{i=1}^{n} (a_i - a_i) g_i(a) = 0.$$

Hence $f \in I(a)$. Therefore

$$(x_1 - a_1, \dots, x_n - a_n) \subseteq I(a).$$

For the reverse, let $f \in I(a)$, so

$$f(a_1, \dots, a_n) = 0.$$

We can write

$$\begin{aligned} f(x_1, \dots, x_n) - f(a_1, \dots, a_n) = {} & (f(x_1, x_2, \dots, x_n) - f(a_1, x_2, \dots, x_n)) \\ & + (f(a_1, x_2, \dots, x_n) - f(a_1, a_2, x_3, \dots, x_n)) \\ & + \cdots \\ & + (f(a_1, \dots, a_{n-1}, x_n) - f(a)). \end{aligned}$$

The first summand is divisible by $x_1 - a_1$, the second by $x_2 - a_2$, and so on. Therefore

$$f - f(a) \in (x_1 - a_1, \dots, x_n - a_n).$$

But $f(a) = 0$, so

$$f \in (x_1 - a_1, \dots, x_n - a_n)$$

giving us the inclusion. Together

$$I(a) = (x_1 - a_1, \dots, x_n - a_n).$$

■

Immediately, the evaluation map

$$\mathrm{ev}_a : K[x_1, \dots, x_n] \to K, \; f \mapsto f(a)$$

has

$$\ker \mathrm{ev}_a = I(a).$$

It is surjective so

$$\frac{K[x_1, \dots, x_n]}{I(a)} \cong K.$$

So far our ideals lived in $K[x_1, \dots, x_n]$. Suppose X is already a variety.

If $Y \subseteq X$, we can define

$$I_X(Y) = \{f \in A(X) \mid f(y) = 0 \text{ for every } y \in Y\}.$$

Conversely, if $S \subseteq A(X)$, define

$$V_X(S) = \{x \in X \mid f(x) = 0 \text{ for every } f \in S\}.$$

We surpress subscripts when the variety is clear.

Proposition 4.3. *If $Y \subseteq Y' \subseteq X$, then*

$$I(Y') \subseteq I(Y).$$

The proof of this is quite short so I won't bother writing it. +

Likewise, if

$$S \subseteq S' \subseteq A(X),$$

then

$$V(S') \subseteq V(S).$$

Intuitively, larger geometry has a direct correspondence with a smaller ideal, and at the extremes

$$\mathbb{A}^n_K = (0)$$

in the usual infinite-field situation, while

$$I(\varnothing) = K[x_1, \dots, x_n] = (1),$$

since every polynomials vacuously vanishes on the empty set.

Suppose $Y \subseteq X$.

We can form

$$Y \mapsto I(Y) \mapsto V(I(Y)).$$

Ceirtainly,

$$Y \subseteq V(I(Y)).$$

Every polynomials in $I(Y)$ was defined precisely to vanish on Y. Similarly, for $S \subseteq A(X)$,

$$S \subseteq I(V(S)).$$

If Y is already a subvariety, something stronger holds.

Proposition 4.4. *If $Y \subseteq X$ is a subvariety, then*

$$V(I(Y)) = Y.$$

Proof. We know $Y \subseteq V(I(Y))$. Because Y is a subvariety, there exists some $S \subseteq A(X)$ such that

$$Y = V(S).$$

Every element of S vanishes on Y, so

$$S \subseteq I(Y).$$

Since taking zero loci reverses inclusion,

$$V(I(Y)) \subseteq V(S) = Y.$$

■

Now for a really nice result.

Let $Y \subseteq X$ be a subvariety.

Any polynomial function on X can be restricted to Y. Thus we get a ring homomorphism

$$\rho A(X) \to A(Y),\ f \mapsto f\big|_Y\ .$$

Proposition 4.5. *The map ρ is surjective and*

$$\ker \rho = I(Y).$$

Hence

$$A(X)/I(X) \cong A(Y).$$

Proof. By the definition of polynomial functions on Y, every polynomial function on Y is obtained by restricting a polynomial function from the ambient space, so ρ is surjective. Furthermore,

$$f \in \ker \rho$$

if and only if

$$f\big|_Y = 0,$$

which means precisely that

$$f(y) = 0$$

for every $y \in Y$.

Thus $\ker \rho = I(Y)$. The first isomorphism theorem then gives

$$A(X)/I(Y) = \frac{K[x_1, \dots, x_n]/I(X)}{I(Y)} \cong A(Y).$$

■

The last major thing now concerns maps between varieties.

Let $X \subseteq \mathbb{A}^n_K$ and $Y \subseteq \mathbb{A}^m_K$ be varieties.

Definition 4.4. A **morphism of varieties**

$$f : X \to Y$$

is a map given by polynomials:

$$f(x) = (f_1(x), \dots, f_m(x)),$$

where

$$f_i \in K[x_1, \dots, x_n].$$

Gathmann then associates to f a ring homomorphism

$$f^* : A(Y) \to A(X)$$

defined by

$$f^*(g) = g \circ f.$$

Notice the direction:

$$X \xrightarrow{f} Y$$

produces

$$A(Y) \xrightarrow{f^*} A(X).$$

The cool thing now is that the arrow reverses.

Suppose you have an ordinary function $f : X \to Y$ and a real-valued function $g : Y \to \mathbb{R}$. Composition gives $g \circ f : X \to \mathbb{R}$.

So a map of spaces $X \to Y$ naturally lets you pull functions backwards

$$\{\text{functions on } Y\} \to \{\text{functions on } X\}.$$

This is called **pullback**.

Hence the notation f^*. This contravariance is very important further into algebraic geometry.

4.1 A few practice problems

Problem 4.1. Determine geometrically

$$V(xy, xz) \subseteq \mathbb{A}^3_K.$$

Try to describe it as a union of familiar linear pieces.

Solution. We have

$$V(xy, xz) = \{(x, y, z) \in K^3 \,:\, xy = 0, xz = 0\}.$$

Both equations share a common factor of x. If $x = 0$, then both equations vanish, so this gives the plane

$$V(x).$$

If $x \neq 0$, then since K is a field we may cancel x to get $y = 0$, $z = 0$, giving us the x-axis $V(y, z)$. Therefore

$$V(xy, xz) = V(x) \cup V(y, z).$$

Geometrically this is the union of the yz-plane and x-axis. ■

Problem 4.2. Let $a = (2, -1) \in \mathbb{A}^2_{\mathbb{R}}$. Without a long proof, determine $I(a)$ and identify

$$\mathbb{R}[x, y]/I(a)$$

Solution. We showed earlier that

$$I(a) = (x - 2, y + 1).$$

Define the ring homomorphism

$$\phi : \mathbb{R}[x, y] \to \mathbb{R}, \; f \mapsto f(2, -1).$$

The map is surjective and $\ker \phi = I(a)$. Thus

$$\mathbb{R}[x, y]/I(a) \cong \mathbb{R}.$$

■

5 Operations on ideals

Let R be a commutative ring with $1 \in R$.

Definition 5.1. Let $f_1, \ldots, f_n \in R$. The ideal generated by $f_1, \ldots, f_n$ is

$$(f_1, \ldots, f_n) = \left\{ \sum_{i=1}^{n} a_i f_i \;\middle|\; a_1, \ldots, a_n \in R \right\}.$$

It is the smallest ideal containing all of

$$f_1, \ldots, f_n.$$

The coefficients are allowed to be arbitrary elements of the ring. For instance, in $K[x, y]$,

$$(x, y) = \{fx + gx \mid f, g \in K[x, y]\},$$

meaning

$$x^4 + x^2y + y^7 \in (x, y).$$

For principal ideals (ideals generated by one elements),

$$(a) = \{ra \mid r \in R\}.$$

If $a \mid b$, then $b = ra$ for some $r \in R$ so

$$b \in (a).$$

In fact,

$$a \mid b \Rightarrow (b) \subseteq (a).$$

5.1 Redundant generators

Suppose $I = (f_1, \dots, f_n)$. If one generator already belongs to the ideal generated by the others, it can (and should) be ommited.

For example,

$$(x, y, xy) = (x, y)$$

because

$$xy = y \cdot x \in (x).$$

5.2 Sum of ideals

Let $I, J \triangleleft R$.

Definition 5.2.

$$I + J = \{a + b \mid a \in I,\ b \in J\}.$$

$I + J$ is the smallest ideal such that I, J are contained within it.

Generally, if

$$I = (f_1, \dots, f_n)$$

and

$$J = (g_1, \dots, g_n),$$

then

$$I + J = (f_1, \dots, f_n, g_1, \dots, g_n).$$

It is prudent to remove redundant generators after concatenating.

For instance,

$$(x^2, xy) + (y) = (x^2, xy, y)$$

but

$$xy \in (y)$$

so we simplify to

$$(x^2, y).$$

5.3 Intersection of ideals

The intersection

$$I \cap J$$

is the set of elements which belong to both I, J, meaning it's just the usual intersection of sets operation. Unlike sums, intersections are often harder to compute directly from generators.

Let $R = K[x, y]$. Suppose $f \in (x) \cap (y)$. Then

$$f = ax,\ a \in R$$

and

$$f = by,\ b \in R.$$

Since x, y are relatively prime in the UFD $K[x, y]$ we have that $xy \mid f$. Therefore $f \in (xy)$ giving us that

$$(x) \cap (y) \subseteq (xy).$$

Conversely, $f = hxy = (hx)y = (hy)x$, so $f \in (x)$ and $f \in (y)$. Thus

$$(xy) \subseteq (x) \cap (y).$$

For monomials m_1, m_2 we have

$$(m_1) \cap (m_2) = (\text{lcm}(m_1, m_2)).$$

5.4 Product of ideals

Definition 5.3. For ideals $I, J \triangleleft R$,

$$IJ$$

is the ideal generated by all products ab with $a \in I$ and $b \in J$. Equivalently,

$$IJ = \left\{ \sum_{k=1}^{n} a_k b_k \;\middle|\; a_k \in I,\ b_k \in J \right\}.$$

It is important that the sums are finite,

$$\{ab : a \in I,\ b \in J\}$$

need not be an ideal.

A nice rule is that if $I = (f_1, \dots, f_r)$ and $J = (g_1, \dots, g_s)$ then

$$IJ = (f_i g_j : i \in [r],\ j \in [s]).$$

For instance

$$(x, y)^2 = (x, y)(x, y) = (x^2, xy, y^2)$$

remembering that $xy = yx$.

Proposition 5.1. *For ideals $I, J \triangleleft R$,*

$$IJ \subseteq I \cap J.$$

Proof. Let $a \in I$ and $b \in J$. Then, since I, J are ideals we have $ab = ba \in I$ and $ab \in J$. Thus $ab \in I \cap J$. Therefore every generator belongs to $I \cap J$ and

$$IJ \subseteq I \cap J.$$

■

For powers of ideals we define it as follows. We define $I^2 = II$, $I^3 = III$ and generally

$$I^n = \prod_{i \in [n]} I = \underbrace{I \cdots I}_{n \text{ factors}}.$$

5.5 Ideal quotient

Definition 5.4. Let $I, J \triangleleft R$. The ideal quotient is

$$I : J = \{r \in R \mid rJ \subseteq I\}.$$

For principal ideal $J = (g)$ we have that

$$rJ \subseteq I$$

precisely when

$$rg \in I.$$

For example if we take $(xy) : (x)$ we want all $r \in R$ such that

$$rx \in (xy),$$

meaning

$$rx = hxy$$

for some $h \in K[x, y]$. Since $K[x, y]$ is a domain we can cancel x to get

$$r = hy.$$

Thus $r \in (y)$. Conversely if

$$r = hy,$$

then $rx = hxy \in (xy)$. So, maybe unsurprisingly,

$$(xy) : (x) = (y).$$

5.6 Quotient by nonprincipal ideal

If $J = (g_1, \dots, g_s)$, then

$$rJ \subseteq I$$

means

$$rg_i \in I$$

for every generator g_i. Thus

$$I : (g_1, \dots, g_s) = \bigcap_{i=1}^{s} (I : (g_i)).$$

5.7 Radical of an ideal

Definition 5.5. For an ideal $I \triangleleft R$,

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 1\}.$$

The radical consists of elements which lie in I for sufficiently large power. Clearly

$$I \subseteq \sqrt{I}.$$

Definition 5.6. An ideal is radical if

$$\sqrt{I} = I.$$

Equivalently,

$$r^n \in I \Rightarrow r \in I.$$

A non-example is (x^2) since $x \notin (x^2)$. Interestingly

$$\sqrt{(x^{100})} = \sqrt{(x^2)} = (x).$$

5.8 Nilpotents

Definition 5.7. An element $a \in R$ is nilpotent if

$$a^n = 0$$

for some $n \geq 1$. The set of nilpotents is thus

$$\sqrt{(0)},$$

usually called the nilradical of R.
A ring R is called reduced if it has no nonzero nilpotent elements. Equivalently,

$$R \text{ reduced} \Leftrightarrow \sqrt{(0)} = (0).$$

Proposition 5.2. *Let $I \triangleleft R$. Then*

$$I \text{ is radical} \Leftrightarrow R/I \text{ is reduced}.$$

Proof. An element $\overline{a} = a + I \in R/I$ is nilpotent if and only if $\overline{a}^n = 0$ for some $n \geq 1$. This is equivalent to $a^n \in I$. Thus R/I has no nonzero nilpotents exactly when $a^n \in I \Rightarrow a \in I$ which is precisely the statement that I is radical. ∎

There are a lot of these links between behaviours of the ideal and the quotient, two of which from a first course in algebra is the correspondence:

$$I \text{ prime} \Leftrightarrow R/I \text{ domain},$$

and

$$I \text{ maximal} \Leftrightarrow R/I \text{ field}.$$

5.9 Geometry of radicals

For an ideal $I \triangleleft K[x_1, \ldots, x_n]$, we have

$$V(I) = V(\sqrt{I}).$$

Proof. Since $I \subseteq \sqrt{I}$, taking zero loci reverses the inclusion

$$V(\sqrt{I}) \subseteq V(I).$$

Conversely let $x \in V(I)$. Take any $f \in \sqrt{I}$. Then $f^m \in I$ for some m. Therefore $f^m(x) = f(x)^m = 0$. Since K is a field we get

$$f(x) = 0.$$

Therefore $x \in V(\sqrt{I})$, meaning

$$V(I) \subseteq V(\sqrt{I}).$$

All in all:

$$V(I) = V(\sqrt{I}).$$

■

Let $Y \subseteq \mathbb{A}^n_K$. Recall that

$$I(Y) = \{f \,:\, f(y) = 0 \ \forall y \in Y\}.$$

Then:

Proposition 5.3. *The ideal $I(Y)$ is radical.*

Proof. Suppose $f^n \in I(Y)$. Then

$$f(y)^n = 0, \ \forall y \in Y.$$

Since K is a field,

$$f(y) = 0.$$

Therefore $f \in I(Y)$, meaning $I(Y)$ is radical. ■

5.10 Other stuff

Proposition 5.4.

$$\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}.$$

Proposition 5.5.

$$V(I + J) = V(I) \cap V(J).$$

Proposition 5.6.

$$V(IJ) = V(I \cap J) = V(I) \cup V(J).$$

Definition 5.8. Ideals I, J are called coprime or comaximal if

$$I + J = R.$$

Proposition 5.7. *If $I + J = R$, then*

$$IJ = I \cap J.$$

Theorem 5.1 (Chinese remainder theorem)**.** *Let $I_1, \dots, I_n \triangleleft R$. Define*

$$\phi : R \to \prod_{i=1}^{n} R/I_i$$

by

$$r \mapsto (r + I_1, \dots, r + I_n).$$

Then

$$\ker \phi = \bigcap_{i=1}^{n} I_i.$$

If I_i are pairwise coprime, then ϕ is surjective. Thus, by the first isomorphism theorem:

$$R/\bigcap_{i=1}^{n} I_i \cong \prod_{i=1}^{n} R/I_i.$$

5.11 Problems

Let $R = K[x, y]$.

Problem 5.1. Compute

$$(x^3, y) + (x, y^4).$$

Solution.

$$(x^3, y) + (x, y^4) = (x^3, y, x, y^4).$$

Since $x^3 \in (x)$ and $y^4 \in y$ it simplifies to (x, y). We could also observe that $(x^n) \subseteq (x)$ for any n (tying it back to radicals). ■

Problem 5.2. Compute

$$(x^2 y) \cap (xy^3).$$

Sol. Suppose $f \in (x^2y) \cap (xy^3)$. Then

$$f = gx^2y = hxy^3$$

Both are divide by x^2y^3 which is lcm(x^2y, xy^3) in R. Thus $f \in (x^2y^3)$, yielding

$$(x^2y) \cap (xy^3) \subseteq (x^2y^3).$$

Now suppose $f \in (xy)$. Then

$$\begin{aligned} f &= ax^2y^3 \\ &= ax(xy^3) \\ &= ay^2(x^2y). \end{aligned}$$

Thus

$$(x^2y) \cap (xy^3) \supseteq (x^2y^3).$$

■

Problem 5.3. Compute

$$(x, y)(x^2, y^2)$$

Sol. Recall that

$$(x, y)(x^2, y^2) = (xx^2, xy^2, yx^2, yy^2) = (x^3, xy^2, yx^2, y^3).$$

■

Problem 5.4. Compute

$$(x^3y^2) : (x^2).$$

Sol. We want to find all $r \in R$ so that $rx^2 \in (x^3y^2)$. Then

$$\begin{aligned} rx^2 &= hx^3y^2 \\ r &= hxy^2. \end{aligned}$$

Therefore

$$(x^3y^2) : (x^2) = (xy^2).$$

■

Problem 5.5. Compute

$$(x^2, xy) : (x).$$

Sol. We want to find $r \in R$ so that

$$rx = hx^2$$

and

$$rx = hxy$$

similtaneously. Proceeding as earlier we get

$$(x) \cap (y) = (xy).$$

■

Problem 5.6. Compute

$$\sqrt{(x^8, y^{10})}.$$

Sol. We want to find all $r \in R$ so that

$$r^n \in (x^8, y^10), \text{ for some } n \geq 1.$$

This means

$$r^n = ax^8 + by^{10}.$$

We claim $(x, y) \subseteq \sqrt{(x^8, y^{10})}$. For x choose $n = 8$ and similarly for y choose $n = 10$. Thus $x, y \in \sqrt{(x^8, y^{10})}$. This yields the inclusion:

$$(x, y) \subseteq \sqrt{(x^8, y^{10})}.$$

Let $f(x, y)$ be some polynomial in $\sqrt{(x^8, y^{10})}$. Then there exists some m so that

$$f^m \in (x^8, y^{10}).$$

We have

$$f^m = ax^8 + by^{10}$$

Since $x, y \equiv 0$ modulo (x, y) we have

$$f^m = 0$$

and since K is a field we then get

$$f(x, y) = 0$$

so

$$f \in (x, y).$$

■

Problem 5.7. Decide whether

$$(x) + (x-1) = K[x].$$

If yes, use it to compute

$$(x) \cap (x-1).$$

Proof. Recall that

$$I = (x) + (x-1) = (x, x-1).$$

Then $x - (x-1) = 1 \in I$. Now for any $f \in K[x]$ we have $f \cdot 1 = f \in I$ so $I = K[x]$.

Since (x) and $(x-1)$ are comaximal we have $(x) \cap (x-1) = (x)(x-1) = (x^2 - x)$. ∎

Problem 5.8 (Set 2, problem 4 (Gathmann 1.8)). Let $I_1, \dots, I_n$ be pairwise coprime ideals in R. Prove that

$$I_1 \cdots I_n = I_1 \cap \cdots \cap I_n.$$

Proof. We have that for $i \neq j$,

$$I_i + I_j = R.$$

$I_1 \cdots I_n \subseteq I_1 \cap \cdots \cap I_n$ is immediate. Now, for $n = 2$, suppose $f \in I_1 \cap I_2$. Then, since $I_1 + I_2 = R$, there exists $a \in I_1,\ b \in I_2$ so that

$$a + b = 1.$$

Multiplying by x we get

$$ax + bx = x.$$

Since $x \in I_2$ and $a \in I_1$ we have $ax \in I_1 I_2$. Likewise, since $x \in I_1$ and $b \in I_2$ we have $bx \in I_1 I_2$. Thus $x = ax + bx \in I_1 I_2$. Hence

$$I_1 I_2 \supseteq I_1 \cap I_2.$$

For the induction step, assume $n > 2$ and that the claim holds for $n-1$ pairwise coprime ideals.

First, we observe that if I_1 is coprime to each I_k for $k = 2, \dots, n$, then $I_1 + (I_2 \cdots I_n) = R$. Indeed, for each $k \geq 2$, choose $a_k \in I_1$ and $b_k \in I_k$ such that $a_k + b_k = 1$. Expanding the product

$$1 = \prod_{k=2}^{n} (a_k + b_k)$$

yields $1 = A + b_2 \cdots b_n$, where $A \in I_1$ (since every term in the expansion except $b_2 \cdots b_n$ contains at least one factor $a_k \in I_1$). Since $b_2 \cdots b_n \in I_2 \cdots I_n$, this shows that $I_1 + (I_2 \cdots I_n) = R$.

Now, let $J = I_2 \cdots I_n$. By the induction hypothesis, $J = I_2 \cap \cdots \cap I_n$. Since I_1 and J are coprime, applying the $n = 2$ base case to I_1 and J yields

$$I_1 \cdots I_n = I_1 J = I_1 \cap J = I_1 \cap (I_2 \cap \cdots \cap I_n) = I_1 \cap \cdots \cap I_n,$$

completing the induction. ■

5.12 Some problems from set 3

Problem 5.9 (Gathmann 1.19)**.**
Let $\phi : R \to R'$ be a ring homomorphism.
We denote I^c to be the inverse image of an ideal or contraction of I.
For $I \triangleleft R$ the ideal generated by the image of $\phi[I]$ is called the image ideal or extension by ϕ, written as I^e if the morphism is clear from context.
Prove:
(c) $(IJ)^e = I^e J^e$ for all ideals I, J of R.
(d) $(I \cap J)^c = I^c \cap J^c$ for all ideals I, J of R'.

Proof of (c). Let I, J be ideals of R. Let $\phi : R \to R'$ be a homomorphism of ring.

We begin by proving the forward inclusion.

Suppose $x \in IJ$, meaning $\phi(x) \in (IJ)^e$. By definition,

$$x = \sum_i \lambda_i x_i$$

where $\lambda_i \in R,\ x_i \in IJ$. Then,

$$\phi\left(\sum_i \lambda_i x_i\right) = \sum_i \phi(\lambda_i)\phi(x_i).$$

For each i, we have that $\phi(x_i) \in IJ$ and $\phi(\lambda_i) \in R'$. Thus each $\lambda(x_i) \in I^e J^e$. Since $I^e J^e$ absorbs $\phi(\lambda_i)$ we get that each $\phi(\lambda_i)\phi(x_i) = \phi(\lambda_i x_i) \in I^e J^e$. As the ideal is closed under addition we get that

$$\phi(x) = \phi\left(\sum_i \lambda_i x_i\right) \in I^e J^e$$

and as x was arbitrarily chosen we finally get

$$(IJ)^e \subseteq I^e J^e.$$

Now for the reverse inclusion.

Suppose $y \in I^e J^e$. Then

$$y = \sum_k u_k v_k,$$

where $u_k \in I^e,\ v_k \in J^e$. Now, because I^e, J^e are generated by $\phi[I],\ \phi[J]$, write

$$u_k = \sum_i r'_{ki}\phi(a_{ki}),\ v_k = \sum_j s'_{ki}\phi(b_{ki}),$$

with $a_{ki} \in I,\ b_{ki} \in J$ and $r'_{ki}, s'_{ki} \in R'$.

Then

$$\begin{aligned} y &= \sum_k \left(\sum_i r'_{ki}\phi(a_{ki})\right)\left(\sum_j s'_{ki}\phi(b_{kj})\right) \\ &= \sum_{k,i,j} r'_{ki}s'_{kj}\phi(a_{ki})\phi(b_{kj}) \\ &= \sum_{k,i,j} r'_{ki}s'_{kj}\phi(a_{ki}b_{kj}). \end{aligned}$$

But

$$a_{ki}b_{kj} \in IJ,$$

so each

$$\phi(a_{ki}b_{kj})$$

is one of the generators of $(IJ)^e$. Hence

$$y \in (IJ)^e.$$

Therefore,

$$I^e J^e \subseteq (IJ)^e.$$

■

Proof of (d). Let $I, J \triangleleft R'$. For any $x \in R$, we have

$$\begin{aligned} x \in (I \cap J)^c &\iff \phi(x) \in I \cap J \\ &\iff \phi(x) \in I \text{ and } \phi(x) \in J \\ &\iff x \in I^c \text{ and } x \in J^c \\ &\iff x \in I^c \cap J^c. \end{aligned}$$

Therefore, $(I \cap J)^c = I^c \cap J^c$. ■

Problem 5.10 (Gathmann 1.22)**.** Let $I \subset J$ be ideals in a ring R. By Lemma 1.21 (in Gathmann's notes), the extension J/I of J by the quotient map $R \mapsto R/I$ is an ideal in R/I. Prove that

$$(R/I)/(J/I) \cong R/J.$$

Proof. Let

$$\psi : R/I \to R/J$$

be defined by

$$\psi(r+I) = r+J.$$

We first check that ψ is well-defined.

Suppose

$$r+I = s+I.$$

Then

$$r-s \in I.$$

Since $I \subset J$, we also have

$$r-s \in J.$$

Hence

$$r+J = s+J.$$

Thus ψ is well-defined. It is clearly a ring homomorphism. Moreover, ψ is surjective since every element of R/J has the form

$$r+J = \psi(r+I)$$

for some $r \in R$. Computing the kernel we have

$$r+I \in \ker\psi$$

if and only if

$$r+J = J.$$

Equivalently,

$$r \in J.$$

Therefore

$$\ker\psi = \{r+I \mid r \in J\} = J/I.$$

Hence, by the First Isomorphism Theorem,

$$(R/I)/(J/I) = (R/I)/\ker\psi \cong \operatorname{im}\psi.$$

As ψ is surjective,

$$(R/I)/(J/I) \cong R/I.$$

■

Problem 5.11. View $\mathbb{Z}$ as a $\mathbb{Z}$-module. Prove that $2\mathbb{Z}$ is a submodule. Then, determine whether $2\mathbb{Z} \cup 3\mathbb{Z}$ is a submodule of $\mathbb{Z}$.

Proof & solution. We have that $2\mathbb{Z} = \{\ldots, -2, 0, 2, \ldots\} \neq \emptyset$. Furthermore take any $x, y \in 2\mathbb{Z}$. Then there are some n, m so that

$$x = 2n,\ y = 2m$$

so

$$x - y = 2n - 2m = 2(n - m) \in 2\mathbb{Z}$$

hence the subgroup criterion is satisfied. Now take $r \in \mathbb{Z}$. Then

$$r \cdot x = r \cdot 2n = 2(rn) \in 2\mathbb{Z}.$$

Thus $2\mathbb{Z}$ is a $\mathbb{Z}$-module and, of course

$$2\mathbb{Z} \subseteq \mathbb{Z}.$$

The latter is not a submodule as it is not even a group:

$$2 + 3 = 5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}.$$

Problem 5.12. Let

$$M = \mathbb{Z}/12\mathbb{Z}$$

viewed as a $\mathbb{Z}$-module, and let

$$N = (\overline{4}).$$

1. Determine N,
2. Determine the elements of the quotient module

$$M/N,$$

3. Find a familiar $\mathbb{Z}$-module isomorphic to M/N.

Sol.

$$(\overline{4}) = \{\overline{0}, \overline{4}, \overline{8}\}.$$

$$M/N = \{m + N \mid m = 0, 1, 2, \dots, 12\} = \{\overline{m} + N \mid \overline{m} = 0, 1, 2, 3\}.$$

Let $\phi : M \to N \to \mathbb{Z}_4$ by $\phi(m) = m \mod 4$. Then $\ker \phi = N$ and it is indeed surjective so the first isomorphism theorem gives us

$$M/N \cong \mathbb{Z}_4.$$

Problem 5.13. View

$$M = \mathbb{Z}/12\mathbb{Z}$$

as a $\mathbb{Z}$-module. Find a composition series

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M.$$

For each i determine the quotient

$$\frac{M_i}{M_{i-1}}.$$

Finally, use this to determine $\ell(M)$.

Proof. We find the series

$$0 \subsetneq \mathbb{Z}/6\mathbb{Z} \subsetneq \mathbb{Z}/2\mathbb{Z} \subsetneq \mathbb{Z}/12\mathbb{Z}.$$

We have

$$\frac{\mathbb{Z}/12\mathbb{Z}}{\mathbb{Z}/2\mathbb{Z}} \cong \mathbb{Z}/2\mathbb{Z},$$

$$\frac{\mathbb{Z}/2\mathbb{Z}}{\mathbb{Z}/6\mathbb{Z}}$$

has size $6/2 = 3$ and since $\mathbb{Z}$-modules are abelian groups we must have that it is isomorphic to $\mathbb{Z}_3$. Lastly

$$\frac{\mathbb{Z}/6\mathbb{Z}}{(0)} \cong \mathbb{Z}/6\mathbb{Z} \cong 2\mathbb{Z}.$$

All of the above are simple so this is a composition series with $\ell(M) = 3$. ■

Problem 5.14. Suppose M has finite length and $N \subseteq M$ is a submodule. Assume you already know that both N and M/N have finite length. Suppose

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_r = N$$

is a composition series for N, and

$$0 = Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_s = M/N$$

is a composition series for M/N. Let

$$\pi : M \to M/N$$

be the quotient map.
Construct a composition series for M using the $N_i's$ inverse images

$$\pi^{-1}[Q_j].$$

Deduce then that

$$\ell(M) = \ell(N) + \ell(M/N).$$

Proof. For each $j \in \{0, 1, \ldots, s\}$, since Q_j is a submodule of M/N, its preimage under the quotient map π, denoted $\pi^{-1}[Q_j]$, is a submodule of M containing $\ker \pi = N$.

Evaluating the preimages at the endpoints of the composition series for M/N, we have:

$$\pi^{-1}[Q_0] = \pi^{-1}[\{0\}] = \ker \pi = N, \quad \text{and} \quad \pi^{-1}[Q_s] = \pi^{-1}[M/N] = M.$$

Furthermore, because $Q_{j-1} \subsetneq Q_j$ is a strict inclusion for each $1 \leq j \leq s$, their preimages must also satisfy strict inclusion:

$$\pi^{-1}[Q_{j-1}] \subsetneq \pi^{-1}[Q_j].$$

Concatenating the composition series of N with the preimages of the composition series of M/N gives the chain of submodules in M:

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_r = N = \pi^{-1}[Q_0] \subsetneq \pi^{-1}[Q_1] \subsetneq \cdots \subsetneq \pi^{-1}[Q_s] = M.$$

To verify that this chain forms a composition series for M, we must check that every successive quotient is simple:

1. For $1 \leq i \leq r$, the quotient N_i/N_{i-1} is simple by the assumption that $(N_i)_{i=0}^r$ is a composition series for N.
2. For $1 \leq j \leq s$, define the map

$$\tilde{\varphi}_j : \pi^{-1}[Q_j] \longrightarrow \frac{Q_j}{Q_{j-1}} \quad \text{by} \quad \tilde{\varphi}_j(m) = \pi(m) + Q_{j-1}.$$

   This is an R-module homomorphism, being the composition of the quotient map π with the canonical projection $Q_j \to Q_j/Q_{j-1}$.

   - **Surjectivity:** Take any element $x + Q_{j-1} \in Q_j/Q_{j-1}$ with $x \in Q_j$. Since π is surjective, there exists $m \in M$ such that $\pi(m) = x$. By definition, $m \in \pi^{-1}[Q_j]$, and $\tilde{\varphi}_j(m) = x + Q_{j-1}$. Thus, $\tilde{\varphi}_j$ is surjective.
   - **Kernel:** An element $m \in \pi^{-1}[Q_j]$ lies in $\ker \tilde{\varphi}_j$ if and only if $\pi(m) + Q_{j-1} = 0 \in Q_j/Q_{j-1}$. This is equivalent to $\pi(m) \in Q_{j-1}$, which means $m \in \pi^{-1}[Q_{j-1}]$. Thus, $\ker \tilde{\varphi}_j = \pi^{-1}[Q_{j-1}]$.

   By the First Isomorphism Theorem, $\tilde{\varphi}_j$ induces an isomorphism of R-modules:

$$\frac{\pi^{-1}[Q_j]}{\pi^{-1}[Q_{j-1}]} \cong \frac{Q_j}{Q_{j-1}}.$$

   Since Q_j/Q_{j-1} is simple by the definition of the composition series for M/N, the factor module $\pi^{-1}[Q_j]/\pi^{-1}[Q_{j-1}]$ is also simple.

Every factor in the spliced chain is simple, so the sequence of submodules

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_r = \pi^{-1}[Q_0] \subsetneq \pi^{-1}[Q_1] \subsetneq \cdots \subsetneq \pi^{-1}[Q_s] = M$$

is a valid composition series for M.

Finally, by the Jordan–Hölder Theorem, all composition series of M have the same length

$\ell(M)$, which is equal to the number of simple factors. Counting the factors in our constructed series gives:

$$\ell(M) = r + s = \ell(N) + \ell(M/N).$$

■

Problem 5.15. Let $M = \mathbb{Z}/36\mathbb{Z}$ as a $\mathbb{Z}$-module and let

$$N = \langle \overline{6} \rangle \leq M.$$

Determine N then identify it up to isomorphism.
Find $\ell(N),\ \ell(M/N),\ \ell(M)$.
Finally check that

$$\ell(M) = \ell(N) + \ell(M/N).$$

Proof. Firstly we have

$$N = \langle \overline{6} \rangle = \{\overline{0}, \overline{6}, \overline{12}, \dots, \overline{30}\},$$

as this is an abelian group, we can use the classification theorem for finitely generated abelian groups to get

$$N \cong \mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6.$$

Now we want to find a maximal submodule of M. Clearly

$$\langle \overline{2} \rangle = \{\overline{0}, \overline{2}, \dots, \overline{34}\}$$

is such a submodule and it is isomorphic to $\mathbb{Z}_{18}$ since it is a cyclic abelian group of order 18.

Lastly if we take $\langle \overline{18} \rangle$ we get a composition chain

$$0 < \langle \overline{18} \rangle < N < \langle \overline{2} \rangle < M.$$

So $\ell(M) = 4$. Now

$$M/N = \frac{\mathbb{Z}/36\mathbb{Z}}{\mathbb{Z}_6} \cong \mathbb{Z}_6$$

so $M/N \cong N$.

$\ell(M/N) = \ell(N) = 2$ since

$$0 < \langle \overline{3} \rangle < \langle \overline{2} \rangle < N = M/N$$

is a valid chain.

Then $4 = \ell(M) = \ell(M/N) + \ell(N) = 2 + 2$ is satisfied. ■

Problem 5.16. Let M be an R-module of finite length, and let

$$f : M \to M$$

be an R-module homomorphism (endomorphism).
Prove that f is injective if and only if f is surjective.
You may use

$$M/\ker f \cong \operatorname{im} f$$

but no assumptions about exact sequences.

Proof. M has finite length so we can find some composition series

$$0 = M_0 < M_1 < \cdots < M_n = M.$$

We are given that

$$M/\ker f \cong \operatorname{im} f.$$

Suppose f is injective. Then $\ker f = \{0\}$. Since

$$M/\ker f \cong \operatorname{im} f$$

we have

$$M = M/\ker f \cong \operatorname{im} f$$

and conversely $M = \operatorname{im} f$ would imply

$$M/\ker f \cong M,$$

but isomorphism doesn't necessarily imply equality.

Suppose first f is injective. Then $\ker f = 0$ so

$$\ell(M) = \ell(0) + \ell(\operatorname{im} f) = \ell(\operatorname{im} f).$$

But $\operatorname{im} f \leq M$. If f were a proper submodule it would have strictly less length. Since their length are equal we have $\operatorname{im} f = M$. So f is surjective.

Conversely, suppose f is surjective. Then $\operatorname{im} f = M$ so we get

$$\ell(M) = \ell(\ker f) + \ell(M)$$

so $\ker f$ has length 0, hence $\ker f = 0$. ■

6 Exact sequences

An exact sequence records, in one line, how kernels, images, submodules, quotients, injections, and surjections fit together.

Gathmann introduces exact sequences as a unified graphical language for module constructions that would otherwise require a lot of repetitive prose.

Definition 6.1. Suppose we have R-modules and R-module homomorphisms

$$M_1 \xrightarrow{\phi_1} M_2 \xrightarrow{\phi_2} M_3.$$

We say that the sequence is exact at M_2 if

$$\operatorname{im} \phi_1 = \ker \phi_2.$$

A longer sequence

$$M_1 \to M_2 \to \cdots \to M_n$$

is exact if this equality holds at every interior module.

So if

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact at B, then

$$\operatorname{im} f = \ker g.$$

In particular,

$$g(f(a)) = 0$$

for every $a \in A$, because

$$\operatorname{im} f \subseteq \ker g.$$

But exactness says something stronger than merely $g \circ f = 0$. It says every element killed by g must have come from A.

6.1 Exactness and the zero module

The zero module 0 is very useful in exact sequences as there is only one homomorphism to or from it.

Consider

$$0 \to M \xrightarrow{\varphi} N.$$

Exactness at M says

$$\operatorname{im}(0 \to M) = \ker \varphi.$$

But

$$\operatorname{im}(0 \to M) = 0.$$

Therefore

$$\ker \varphi = 0.$$

So φ is injective.

Thus

$$0 \to M \xrightarrow{\varphi} N$$

is exact if and only if φ is injective.

Similarly,

$$M \xrightarrow{\varphi} N \to 0$$

is exact if and only if $\operatorname{im} \varphi = N$ which is that φ is surjective.

Thus exactness of

$$0 \to M \xrightarrow{\varphi} N \to 0$$

holds if and only if φ is an isomorphism.

6.2 Short exact sequences

The most important exact sequences are the short exact sequences

$$0 \to M \xrightarrow{\varphi} N \xrightarrow{\psi} P \to 0.$$

Exactness tells us three things:

$$\varphi \text{ is injective,}$$
$$\operatorname{im} \varphi = \ker \psi,$$

and

$$\psi \text{ is surjective.}$$

This has a rather nice interpretation.

Because φ is injective, we know

$$M \subseteq N.$$

Because

$$\ker \psi = \operatorname{im} \varphi \cong M,$$

the map ψ kills exactly the copy of M inside N.

And because ψ is surjective,

$$P \cong N / \ker \psi.$$

Therefore

$$P \cong N/M.$$

So conceptually, a short exact sequence

$$0 \to M \to N \to P \to 0$$

means N contains a copy of M, and after quotienting N by that copy, what remains is P.

6.3 The canonical short exact sequence

Suppose $N \leq M$ is a submodule.

Then there is always a short exact sequence

$$0 \to N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \to 0.$$

Here $\iota(n) = n$ is the inclusin map and

$$\pi(m) = m + N$$

is the quotient map.

The inclusion is injective, the quotient map is surjective and

$$\ker \pi = N.$$

Since $\operatorname{im}(\iota) = N$ we have

$$\operatorname{im} \iota = \ker \pi.$$

6.4 A homomorphism produces exact sequences

Take any module homomorphism $\varphi : M \to N$. We always have

$$0 \to \ker \varphi \to M \xrightarrow{\varphi} \operatorname{im} \varphi \to 0.$$

And also

$$0 \to \operatorname{im} \varphi \to N \to N/\operatorname{im} \varphi \to 0.$$

Gluing them gives

$$0 \to \ker \varphi \to M \xrightarrow{\varphi} N \to N/\operatorname{im} \varphi \to 0.$$

The last module

$$N/\operatorname{im} \varphi$$

is called the cokernel of φ:

$$\operatorname{coker} \varphi := N/\operatorname{im} \varphi.$$

So another very useful way to package a map is

$$0 \to \ker \varphi \to M \xrightarrow{\varphi} N \to \operatorname{coker} \to 0.$$

This tells us that the map has two kinds of failure.

$\ker \varphi$ measures failure of injectivity, while $\operatorname{coker} \varphi$ measures failure of surjectivity.

6.5 Splitting and gluing exact sequences

Suppose

$$M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \xrightarrow{\varphi_3} M_4$$

is exact.

Then at M_3,

$$\operatorname{im} \varphi_2 = \ker \varphi_3.$$

Let $N := \operatorname{im} \varphi_2 = \ker \varphi_3$.

Then we can split the sequence into

$$M_1 \to M_2 \to N \to 0$$

and

$$0 \to N \to M_3 \to M_4.$$

Conversely, those pieces can be glued back together.

6.6 Exactness and length

If

$$0 \to M \to N \to P \to 0$$

is exact and all modules have finite length, then

$$\ell(N) = \ell(M) + \ell(P).$$

For a longer exact sequence

$$0 \to M_1 \to M_2 \to \cdots \to M_n \to 0,$$

Gathmann derives the alternating-sum formula

$$\sum_{i=1}^{n} (-1)^i \ell(M_i) = 0.$$

6.7 Can the middle module be determined

Suppose

$$0 \to M \to N \to P \to 0.$$

A tempting guess is

$$N \overset{?}{\cong} M \oplus P.$$

This is sadly not always true.

There is ceirtanly always the obvious example

$$0 \to M \to M \oplus P \to P \to 0,$$

where the first map is the inclusion $m \mapsto (m, 0)$ and the second is the projection $(m, p) \mapsto p$.

Gathmann produces a counter-example to this idea.

6.8 Split exact sequences

A short exact sequence

$$0 \to M \xrightarrow{\varphi} N \xrightarrow{\psi} P \to 0$$

is called split exact when

$$N \cong M \oplus P$$

in a way compatible witht the maps.

Gathmann's Splitting Lemma says the following conditions are equivalent:

$$N \cong M \oplus P,$$

φ has a left inverse

$$\alpha : N \to M,\ \alpha \circ \varphi = \mathrm{id}_M,$$

or ψ has a right inverse

$$\beta : P \to N,\ \psi \circ \beta = \mathrm{id}_P .$$

The map $\beta : P \to N$ is often called a section.

It means that we can choose, in a homomorphic way, one representative in N for every element of P.

Then N decomposes into the kernel part of M plus a copy of P.

For vector spaces, an exact sequence always splits.

6.9 Commutative diagrams

Definition 6.2. A commutative diagram is a diagram of modules and maps in which every path from one objects to another gives the same map.

For example,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle h} \\ C & \xrightarrow[k]{} & D \end{array}$$

is commutative when

$$h \circ f = k \circ g.$$

Suppose we have two exact rows and vertical homomorphisms

$$\begin{array}{ccccccc} M & \longrightarrow & N & \longrightarrow & P & \longrightarrow & 0 \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & M' & \longrightarrow & N' & \longrightarrow & P' \end{array}$$

with the squares commutative.

The Snake Lemma manufactures a new exact sequence involving the kernels and cokernels of the vertical maps:

$$\ker \alpha \to \ker \beta \to \ker \gamma \xrightarrow{\delta} M' / \operatorname{im} \alpha \to N' / \operatorname{im} \beta \to P' \operatorname{im} \gamma.$$

In the notation introduced earlier we write

$$\ker \alpha \to \ker \beta \to \ker \gamma \xrightarrow{\delta} \operatorname{coker} \alpha \to \operatorname{coker} \beta \to \operatorname{coker} \gamma.$$

The map $\delta : \ker \gamma \to \operatorname{coker} \alpha$ is called the connecting homomorphism.

Conceptually, if you have two exact rows linked by vertical maps, the Snake Lemma relates the failures of those vertical maps to be injective and surjective.

Kernels measure failure of injectivity, cokernels measure the failure of surjectivity. The Snake Lemma ties them nicely together.

Gathmann constructs γ by doing a diagram chase: life an element backwards through one map, move vertically and life backwards again.

6.10 Some problems to see if I understand what the hell is going on

Problem 6.1. Let

$$A \xrightarrow{f} B \xrightarrow{g} C$$

be a sequence of R-modules. Suppose

$$\operatorname{im} f = \ker g.$$

Answer the following:

1. At which module is the sequence exact?
2. Must f be injective?
3. Must g be surjective?
4. Prove that
$$g \circ f = 0.$$

Solution and proof sketch. The sequence is exact at B.

f needs not be injective and g needs not be surjective.

We have $\operatorname{im} f = \ker g$. Then $g(x) = 0$ for all $x = f(a) \in \operatorname{im} f$. Then $g \circ f = 0$. ■

Problem 6.2. Let

$$0 \to M \xrightarrow{\varphi} N$$

be exact.
Prove directly from the definition that φ is injective.
Then let

$$M \xrightarrow{\psi} N \to 0$$

be exact and prove that ψ is surjective.
Finally explain why

$$0 \to M \xrightarrow{\alpha} N \to 0$$

is exact if and only if α is an isomorphism.

Proof sketch. There exists precisely one homomorphism from $0 \to M$, that being the 0-homomorphism. Thus $\operatorname{im}(0 \to M) = \{0\}$. Then, if the sequence is exact, then $\{0\} = \operatorname{im}(0 \to M) = \ker \varphi$ so φ is injective.

Likewise, there exists one and only one homomorphism from N to 0, that being the 0-homomorphism. Thus, if $\operatorname{im} \psi = \ker(N \to 0) = N$ so ψ is surjective.

Summing up the previous two it is easy to see that we require $\ker \alpha = \{0\}$ and $\operatorname{im} \alpha = N$ so it is bijective and thus an isomorphism.

Suppose now that α is an isomorphism. Then $\ker \alpha = \{0\}$ and $\operatorname{im} \alpha = N$ which, together

with the fact that only the 0-homomorphism is valid from $0 \to M$ and $N \to 0$ gives the equivalence we want. ■

Problem 6.3. Let $N \leq M$ be a submodule. Consider

$$0 \to N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \to 0,$$

where $\iota(n) = n$ and $\pi(m) = m + N$. Prove this sequence is exact.

Proof. Trivially, $\operatorname{im}(0 \to N) = \{0\}$ and $\ker(M/N \to 0) = M/N$.

If $\ker \iota \supsetneq \{0\}$ then there would be nonzero $n \in M$ such that $n = 0$ which would imply $M = N = 0$, which in turn would imply $\ker \iota = \{0\}$. (This was a really dumb argument, but it is obvious that $\ker \iota = 0$).

Next, π is surjective ($\operatorname{im} \pi = M/N$) as we have seen before.

It remains to show that $\operatorname{im} \iota = \ker \pi$.

Suppose $\pi m = 0$. Then $\overline{m} + N = 0 + N$ so $m \in N$ so $\ker \iota = N$.

$\operatorname{im} \iota = N$ is trivial to see. ■

Problem 6.4. Consider

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\psi} \mathbb{Z}/2\mathbb{Z} \to 0,$$

where $\varphi(\overline{1}) = \overline{2}$ and ψ is reduction modulo 2. Prove it is exact.
Decide whether it is split exact.

Proof. We have $\ker \varphi = \overline{0} = \operatorname{im}(0 \to \mathbb{Z}/2\mathbb{Z})$. Next, $\operatorname{im} \varphi = \{\overline{0}, \overline{2}\} = \ker \psi$ since ψ is reduction modulo 2. Lastly $\operatorname{im} \psi = \mathbb{Z}/2\mathbb{Z} = \ker(\mathbb{Z}/2\mathbb{Z} \to 0)$.

This sequence is not split exact because $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \ncong \mathbb{Z}/4\mathbb{Z}$. (The latter has an element of order 4, the former does not). ■

Problem 6.5. Let $I, J \triangleleft R$ be ideals. Prove there is a short exact sequence

$$0 \to I \cap J \to I \oplus J \to I + J \to 0.$$

Proof. Consider $\varphi : I \cap J \to I \oplus J$ and $\psi : I \oplus J \to I + J$ by

$$\varphi(x) = (x, -x),$$

and

$$\psi(x, y) = x + y.$$

$\operatorname{im}(0 \to I \cap J) = \{0\}$ and $\ker \varphi = \{0\}$. Also $\operatorname{im} \varphi = \{(n, n) \mid n \in I \text{ and } n \in J\}$. Then it is easy to see that $\ker \psi = \operatorname{im} \varphi$. The final correspondence is similarly easy. ■

Problem 6.6. Let $R = \mathbb{Z}$ with ideal $I = (5)$. Let $M = \mathbb{Z}_{12}$. Show that $M = IM$. Can you find an element $a \in I$ such that $am = m$ for every $m \in M$.

Solution and proof. Recall that IM is the set of all finite sums $\sum a_i m_i$ where $a_i \in I,\ m_i \in M$.

First we find an $a \in I$ so that $am = m$. Notice that $a := 25 \in I$ is 1 modulo 12 so

$$am \equiv 1 \cdot m = m.$$

This means that for every $m \in M$ we have

$$m = 25m \in IM$$

so $M \subseteq IM$. The reverse inclusion is automatic from the definition of an R-module. ■

7 Tensor Products

We have covered machinery for modules and linear maps, but many natural maps are not linear in one variable. Rather, they are bilinear.

For instance if M, N, P are R-modules, a map

$$\alpha : M \times N \to P$$

is R-bilinear if, for fixed n, the map

$$m \mapsto \alpha(m, n)$$

is linear, and for fixed m, the map

$$n \mapsto \alpha(m, n)$$

is linear.

We want to build a new module $M \otimes_R N$ such that studying bilinear maps out of $M \times N$ is equivalent to studying ordinary linear maps out of $M \otimes_R N$.

More precisely, we want

$$\{\text{bilinear maps } M \times N \to P\} \leftrightarrow \{\text{linear maps } M \otimes_R N \to P\}.$$

7.1 Motivation from free modules

Suppose first that $M = R^m$ and $N = R^n$ with bases

$$b_1, \dots, b_m$$

and

$$c_1, \dots, c_n.$$

A bilinear map

$$\alpha : M \times N \to P$$

is determined by the mn values

$$\alpha(b_i, c_j).$$

Indeed

$$\alpha\left(\sum_i \lambda_i b_i, \sum_j \mu_j c_j\right) = \sum_{i,j} \lambda_i \mu_j \alpha(b_i, c_j).$$

We imagine constructing a free module with basis symbols $b_i \otimes c_j$. Call it F. Then a linear map

$$F \to P$$

is also determined by choosing arbitrary values for those mn basis elements. Thus

$$\alpha(b_i, c_j) \leftrightarrow \varphi(b_i \otimes c_j).$$

So for free modules it should be natural that

$$R^m \otimes_R R^n \cong R^{mn}.$$

7.2 The universal property

Definition 7.1 (Tensor product). A **tensor product** of M and N over R consists of an R-module

$$M \otimes_R N$$

together with a bilinear map

$$\tau : M \times N \to M \otimes_R N,\ (m, n) \mapsto m \otimes N,$$

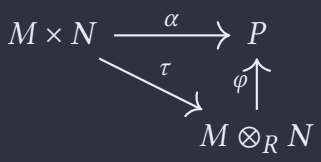
such that every bilinear map

$$\alpha : M \times N \to P$$

factors uniquely through a linear map

$$\varphi : M \otimes_R N \to P.$$

Diagramatically,



with

$$\alpha = \varphi \circ \tau.$$

This is called the universal property.

Once we know what α should do, the universal product gives us a unique linear map from the tensor product.

This definition tells us what it does rather than giving a construction. This is sufficient because the definition uniquely determines the tensor product up to isomorphism. So if two modules T_1, T_2 perform this job then there is a unique isomorphism $T_1 \cong T_2$ comptaible with their tensor maps.

The symbol $m \otimes n$ is not an ordered pair. It denotes an element of the new module $M \otimes_R N$. The defining bilinearity gives the relations

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$$

and

$$m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2,$$

and

$$(am) \otimes n = a(m \otimes n) = m \otimes (an).$$

7.3 Pure tensors vs. global tensors

An element specifically of the form $m \otimes n$ is called a pure tensor. But not every element of $M \otimes_R N$ is of this form. A general tensor is a finite sum

$$\sum_{i=1}^{k} a_i (m_i \otimes n_i).$$

Pure tensors generate the tensor product, but don't necessarily form a basis, and representations are genrally not unique.

7.4 The construction

Start with a large free R-module F whose basis is indexed by all pairs

$$(m, n) \in M \times N.$$

At this stage,

$$(m_1 + m_2, n)$$

and

$$(m_1, n) + (m_2, n)$$

are totally unrelated formal expressions. We want them to become equal, because tensoring should be bilinear. Take a submodule G generated by all expressions forcing bilinearity

$$(m_1 + m_2, n) - (m_1, n) - (m_2, n),$$

$$(am, n) - a(m, n),$$

$$(m, n_1 + n_2) - (m, n_1) - (m, n_2),$$

$$(m, an) - a(m, n).$$

Then define

$$M \otimes_R N := F/G.$$

7.5 Three isomorphisms

First,

$$M \otimes_R N \cong N \otimes_R M.$$

The map is $m \otimes n \mapsto n \otimes m$.

Second,

$$M \otimes_R R \cong M.$$

The map is $m \otimes a \mapsto am$. The inverse map is $m \mapsto m \otimes 1$.

Third, tensor products distribute over direct sums

$$(M \oplus N) \otimes_R P \cong (M \otimes_R P) \oplus (N \otimes_R P).$$

Also, tensor products are associative up to natural isomorphism

$$(M \otimes N) \otimes P = M \otimes (N \otimes P)$$

so we usually just write

$$M \otimes N \otimes P.$$

7.6 Free modules

If $M \cong R^m$ and $N \cong R^n$ then

$$M \otimes N \cong R^{mn}.$$

If $b_1, \ldots, b_m$, $c_1, \ldots, c_n$ are bases, then

$$b_i \otimes c_j$$

forms a basis of the tensor product. So over a field,

$$\dim(V \otimes W) = \dim V \cdot \dim W.$$

7.7 Vanishing

Suppose $I, J \triangleleft R$ are coprime ideals, meaning

$$I + J = R.$$

Then

$$R/I \otimes_R R/J = 0.$$

Choose $a \in I, b \in J$ with $a + b = 1$. Then, for any pure tensor

$$\bar{r} \otimes \bar{s} = (a + b)(\bar{r} \otimes \bar{s}) = a\bar{r} \otimes \bar{s} + \bar{r} \otimes b\bar{s}.$$

But $a\bar{r} = 0$ in R/I and $b\bar{s} = 0$ in R/J. Therefore $\bar{r} \otimes \bar{s} = 0$.

So for distinct primes p, q we have

$$\mathbb{Z}/p\mathbb{Z} \otimes \mathbb{Z}/q\mathbb{Z} = 0.$$

Sidenote:

In general

$$M' \subseteq M \nRightarrow M' \otimes N \subseteq M \otimes N.$$

7.8 Tensor product of homomorphisms

Suppose $\varphi : M \to N$ and $\varphi' : M' \to N'$ are module homomorphisms. Then there is a natural homomorphism

$$\varphi \otimes \varphi' : M \otimes M' \to N \otimes N'$$

defined on pure tensors by $(\varphi \otimes \varphi')(m \otimes m') = \varphi(m) \otimes \varphi(m')$.

This is well-defined because the map

$$(m, m') \mapsto \varphi(m) \otimes \varphi(m')$$

is bilinear, so the universal property gives the corresponding linear map.

7.9 Extension of scalars

Suppose R' is an R-algebra and M is an R-module.

We can turn M into an R'-module by defininining

$$M_{R'} := M \otimes_R R'.$$

This is called an extension of scalars. The new scalar multiplication is

$$a \cdot (m \otimes s) = m \otimes (as), \ a, s \in R'.$$

As a motivating example, consider a real vector space V. To allow complex coefficients, define

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

This is the complexification of V. If $\dim_{\mathbb{R}} V = n$ then

$$V_{\mathbb{C}} \cong \mathbb{C}^n.$$

7.10 Tensor products of algebras

If R_1, R_2 are not merely R-modules, but R-algebras, then

$$R_1 \otimes R_2$$

itself is an R-algebra.

Define multiplication by pure tensors as

$$(s \otimes t)(s' \otimes t') = ss' \otimes tt'.$$

Bilinearity extends to arbitrary tensors. A nice example is

$$R[x, y] \cong R[x] \otimes_R R[y].$$

The map is

$$f(x) \otimes g(x) = f(x)g(x).$$

In particular,

$$x^i \otimes y^j = x^i y^j.$$

7.11 The Hom-tensor relation

There is a natural isomorphism

$$\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)) \cong \mathrm{Hom}_R(M \otimes_R N, P).$$

Suppose $\alpha : M \to \mathrm{Hom}(N, P)$. For each m, $\alpha(m)$ is itself a map $N \to P$, so we can define

$$\beta(m \otimes n) = \alpha(m)(n).$$

Conversely, given

$$\beta : M \otimes N \to P,$$

define

$$\alpha(m)(n) = \beta(m \otimes n).$$

7.12 Tensor products are right exact

Suppose

$$M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \to 0$$

is exact. Tensor everything with N,

$$M_1 \otimes N \xrightarrow{\varphi_1 \otimes \mathrm{id}} M_2 \otimes N \xrightarrow{\varphi_2 \otimes \mathrm{id}} M_3 \otimes N \to 0.$$

Then this sequence is exact.

Interestingly, tensor products are not left exact.

7.13 Some practice problem

Problem 7.1. Work over $\mathbb{Z}$. Simplify as far as possible:

$$(3a + 2b) \otimes n,$$

and show directly from the tensor rules that

$$6 \otimes \overline{1} = 0$$

in

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z}.$$

Then decide whether

$$2 \otimes \overline{1}$$

is zero in the same tensor products.

Solution. We have

$$\begin{aligned}(3a + 2b) \otimes n &= 3a \otimes n + 2b \otimes n \\ &= 3(a \otimes n) + 2(b \otimes n).\end{aligned}$$

We have

$$\begin{aligned}6 \otimes \overline{1} &= 2 \cdot 3 \otimes \overline{1} \\ &= 1 \otimes (2 \cdot 3)\overline{1} \\ &= 1 \otimes (2)\overline{0} \\ &= 1 \otimes \overline{0} \\ &= 0\end{aligned}$$

in $\mathbb{Z} \otimes \mathbb{Z}/3\mathbb{Z}$.

No we do not have $2 \otimes \overline{1} = 0$.

We have

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z}$$

via $a \times \overline{b} \mapsto \overline{ab}$ so we get

$$2 \otimes \overline{1} \mapsto \overline{2} \neq \overline{0}.$$

Problem 7.2. Define

$$\alpha : k[x] \times k[y] \to k[x, y],\ \alpha(f, g) = fg.$$

Prove α is bilinear. Use the universal property to obtain a linear map

$$\varphi : k[x] \otimes_k k[y] \to k[x, y]$$

satisftying

$$\varphi(f \otimes g) = fg.$$

Finally argue that φ is an isomorphism by looking at the elements

$$x^i \otimes y^j.$$

Proof. Fix g. Then

$$\begin{aligned} \alpha(f + h, g) &= (f + h)g \\ &= fg + hg \\ &= \alpha(f, g) + \alpha(h, g) \end{aligned}$$

and

$$\begin{aligned} a\alpha(f, g) &= \alpha(af, g) \\ &= (af)g \\ &= f(ag) \\ &= \alpha(f, ag) \end{aligned}$$

so for fixed g this is linear. The procedure is the same for showing linearity in the second argument.

For the tensor product we have a map $\tau : k[x] \times k[y] \to k[x] \otimes_k k[y]$ for every bilinear map, such as $\alpha : k[x] \times k[y] \to k[x, y]$, so that it factors uniquely through a linear map

$\varphi : k[x] \otimes_k k[y] \to k[x,y]$ such that

$$\begin{array}{ccc} k[x]\times k[y] & \xrightarrow{\alpha} & k[x,y] \\ & \searrow^{\tau} & \uparrow{\scriptstyle \varphi} \\ & & k[x]\otimes_k k[y] \end{array}$$

commutes, meaning $\alpha = \varphi \circ \tau$. Recalling that $\alpha(f,g) = fg$, and $\tau(f,g) = f \otimes g$, we know that $\varphi(f \otimes g) = fg$.

The basis for $k[x] \otimes k[y]$ as an R-algebra is $x^i \otimes y^j$ for $i,j \geq 0$ and the basis for $k[x,y]$ is

$$\{1, x, y, x^2y, xy^2, \ldots\}$$

so since elements in each can be written in these bases we have the correspondence:

$$\varphi\left(\sum_{i,j} a_{ij} x^i \otimes y^j\right) = \sum_{i,j} a_{ij} x^i b^j$$

both with finitely many a_{ij} nonzero. ■

Problem 7.3. Prove that

$$M \otimes_R R \cong M$$

from the universal property.

Proof. Let $\alpha : M \times R \to M$ by

$$\alpha(m,r) = rm.$$

This is a well defined, bilinear homomorphism for obvious reasons (trust me bro).

Thus, by the universal property, there is a $\varphi : M \otimes_R R \to M$ making

$$\begin{array}{ccc} M\times R & \xrightarrow{\alpha} & M \\ & \searrow^{\tau} & \uparrow{\scriptstyle \varphi} \\ & & M\otimes_R R \end{array}$$

commute, where $(m,r) \overset{\tau}{\mapsto} m \otimes r$.

So φ is the homomorphism sending $m \otimes r$ to rm.

Let $\psi : M \to M \otimes_R R$ by

$$\psi(m) = m \otimes 1.$$

Clearly, this is a homomorphism. For the isomorphism argument observe the following.

Then $(\varphi \circ \psi)(m) = m$. For the other direction we look at pure tensors:

$$(\psi \circ \varphi)(m \otimes r) = \psi(rm) = rm \otimes 1 = m \otimes r.$$

Thus

$$M \otimes_R R \cong M.$$

■

Problem 7.4. Consider

$$\mathbb{Z}/4\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/6\mathbb{Z}.$$

Define $\alpha : \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ by

$$\alpha(\overline{a}, \overline{b}) = \overline{ab} \mod 2.$$

Show α is well-defined, $\mathbb{Z}$-bilinear, derive a homomorphism from the tensor product into $\mathbb{Z}/2\mathbb{Z}$ and finally prove that

$$\overline{1} \otimes \overline{1} \neq 0.$$

Proof. Clearly, α is well-defined since taking the product of some integers a, b and reducing modulo 2 yields a value in $\mathbb{Z}/2\mathbb{Z}$.

Next we check bilinearity.

$$\begin{aligned}
\alpha(\overline{a_1} + \overline{a_b}, b) &= \overline{(a_1 + a_2)b} \mod 2 \\
&= \overline{a_1 b + a_2 b} \mod 2 \\
&= \overline{a_1 b} + \overline{a_2 b} \mod 2 \\
&= \alpha(a_1, b) + \alpha(a_2, b).
\end{aligned}$$

The check for linearity in the second argument is analogous.

Let τ be the map which sends $(\overline{a}, \overline{b}) \overset{\tau}{\mapsto} \overline{a} \otimes \overline{b}$.

Then there exists $\varphi : \mathbb{Z}/4\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ such that

$$\begin{array}{ccc}
\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} & \xrightarrow{\alpha} & \mathbb{Z}/2\mathbb{Z} \\
& \searrow^{\tau} & \uparrow{\scriptstyle \varphi} \\
& & \mathbb{Z}/4\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/6\mathbb{Z}
\end{array}$$

commutes.

Then φ is the homomorphism sending $\overline{a} \otimes \overline{b} \overset{\varphi}{\mapsto} \overline{ab}$ reduced modulo 2.

Then

$$\begin{aligned}
\varphi(\overline{1} \otimes \overline{1}) &= \alpha(\overline{1}, \overline{1}) \\
\varphi(\overline{1} \otimes \overline{1}) &= \overline{1 \cdot 1} \mod 2 \\
\varphi(\overline{1} \otimes \overline{1}) &= \overline{1} \mod 2 \\
\varphi(\overline{1} \otimes \overline{1}) &\neq 0.
\end{aligned}$$

8 Localization

We begin by motivating localization with something familiar.

We know $\mathbb{Z} \subseteq \mathbb{Q}$. To get from the integers to the rationals we allow division by all nonzero elements of the former.

Localization can be seen as a generalization of this. Given a ring R, choose a set $S \subseteq R$ whose elemennts you want to make invertible.

Then construct a new ring,

$$S^{-1}R$$

whose elements look like fractions a/s with $a \in R,\ s \in S$.

Gathmann gives this algebraic motivation as well as similtaneously giving a geometric interpretation in terms of rational functions defined near a point.

The important point is that we are now interested in a structure $S^{-1}R \approx$ the ring obtained from R by making every $s \in S$ invertible.

8.1 S must be multiplicatively closed

Suppose we want fractions

$$\frac{a}{s}, \frac{b}{t}.$$

Then their product should be

$$\frac{ab}{st}.$$

So if s, t are allowed denominators, st must also be an allowed denominator.

Thus S must be multiplicatively closed, meaning $1 \in S$, $s, t \in S \Rightarrow st \in S$.

The first condition ensures that the original ring embeds, at least superficially, by $a \mapsto a/1$.

8.2 Recognizing same fractions

If R were an integral domain, we'd expect

$$\frac{a}{s} = \frac{a'}{s'}$$

if and only if

$$as' = a's.$$

However, rings can have zero divisors, so we modify this.

Define $(a, s) \sim (a', s')$ if there exists $u \in S$ such that

$$u(as' - a's) = 0.$$

The fraction notation $\frac{a}{s}$ then means the equivalence class (a, s).

The point of the extra u is, because when zero divisors exist, ordinary cross-multiplication does not give a transitive equivalence relation. The extra u roughly says that the two fractions become equal after multiplying by some denominator that we are going to make invertible anyways.

If S contains no zero divisors, this reduces to the typical $as' = a's$.

The resulting ring is

$$S^{-1}R = \left\{ \frac{a}{s} \;\middle|\; a \in R,\ s \in S \right\}.$$

Addition and multiplication behave exactly how you'd expect.

8.3 Geometrically

Suppose $R = A(X)$ is the coordinate ring of an affine variety X, and fix a point $p \in X$. A fraction

$$\frac{f}{g}$$

might not define a function on all of X, since g might vanish somewhere, but if $g(p) \neq 0$, then g is nonzero at the point we care about, so f/g makes sense at least around p.

Thus choose

$$S = \{g \in A(X) \mid g(p) \neq 0\}.$$

Then $S^{-1}A(X)$ is the ring of functions that are defined locally around p.

8.4 Important choices of S

There are several standard localizations, but two are much more important than others.

First, if R is an integral domain, take

$$S = R \setminus \{0\}.$$

Then every nonzero element becomes invertible, so

$$S^{-1}R$$

is a field, namely the field of fractions, often denoted

$$\mathrm{Quot}(R).$$

Obviously,

$$\mathrm{Quot}(\mathbb{Z}) = \mathbb{Q}.$$

Second, fix an element $a \in R$ and take $S = \{1, a, a^2, \ldots\}$. Then we invert a and write the resulting localization as R_a.

Importantly, if P is a prime ideal, take $S = R \setminus P$. Because P is prime, $R \setminus P$ is multiplicatively closed. We write R_P for this localization.

Thus in R_P, every element outside P becomes invertible.

Concretely, localizing $\mathbb{Z}$ at the prime ideal (p) gives

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \;\middle|\; p \nmid b \right\}.$$

So denominators are allowed precisely when they are not divisible by p. For instance,

$$\frac{1}{7} \in \mathbb{Z}_{(3)},$$

but

$$\frac{1}{3} \notin \mathbb{Z}_{(3)}.$$

8.5 Prime ideals

Prime ideals of $S^{-1}R$ correspond exactly to prime ideals of R that do not meet S, meaning the prime ideals of $S^{-1}R$ correspond one-to-one with the prime ideals P of R satisfying $P \cap S = \emptyset$.

Now let $S = R \setminus P$. The primes surviving localization are precisely primes Q satisfying $Q \subseteq P$.

In particular, P becomes the unique largest surviving prime. Consequently, R_P has exactly one maximal ideal.

A ring with exactly one maximal ideal is called a local ring. Thus R_P is a local ring.

The unique maximal ideal is

$$PR_P = \left\{ \frac{a}{s} \;\middle|\; a \in P; s \notin P \right\}.$$

Geometrically, after zooming all the way in around one point, that point becomes the only maximal ideal left.

8.6 The universal property (yippee, another one)

Localization has a universal property.

Suppose $\alpha : R \to R'$ is a ring homomorphism such that every element of S becomes a

unit in A'. Then there exists a unique map

$$\varphi : S^{-1}R \to R'$$

extending α.

This is inevitable because once s becomes invertible, there is only one sensible possible definition, namely

$$\varphi\left(\frac{a}{s}\right) = \alpha(a)\alpha(s)^{-1}.$$

So $S^{-1}R$ is the universal ring in which every element of S has been made invertible.

This is analogous to the tensor product,

$$M \otimes N$$

is the universal object for turning bilinear maps into linear maps while

$$S^{-1}R$$

is the universal object for making the elements of S inveritble.

8.7 Localization of modules

If M is an R-module, we can localizie it as well. Its elements look like

$$\frac{m}{s},\ m \in M,\ s \in S.$$

We denote this module by $S^{-1}M$. The equivalence relation is the obvious module analogue

$$\frac{m}{s} = \frac{m'}{s'}$$

if some $u \in S$ satisfies $u(s'm - sm') = 0$.

Then

$$S^{-1}M$$

is naturally a module over $S^{-1}R$.

If P is prime, we write M_P. Geometrically we can think of this as retaining the behavior of M only near P.

8.8 Tensor products again

Gathmann proves that

$$S^{-1}M \cong M \otimes_R S^{-1}R.$$

So localization of a module is just extension of scalars

$$R \longrightarrow S^{-1}R.$$

We had an R-module M. We enlarge the scalar ring to $S^{-1}R$. The corresponding module becomes

$$M \otimes_R S^{-1}R.$$

The correspondence is nice also:

$$\frac{m}{s} \leftrightarrow m \otimes \frac{1}{s}.$$

8.9 Localization is exact

Suppose

$$0 \to M_1 \xrightarrow{\varphi} M_2 \xrightarrow{\psi} M_3 \to 0$$

is short exact.

Then localization gives an exact sequence

$$0 \to S^{-1}M_1 \to S^{-1}M_2 \to S^{-1}M_3 \to 0.$$

Nicely, this is stronger than tensoring since the tensor product is onkly right exact in general.

But localiztaion is tensoring with $S^{-1}R$, and for this special tensor factor injectivity is always preserved so

$$- \otimes_R S^{-1}R$$

is actually exact.

Later terminology would say that $S^{-1}R$ is a flat R-module.

Immediate consequences include

$$\begin{aligned} S^{-1}(\ker \varphi) &= \ker(S^{-1}\varphi), \\ S^{-1}(\operatorname{im} \varphi) &= \operatorname{im}(S^{-1}\varphi), \\ S^{-1}(N/M) &\cong S^{-1}N/S^{-1}M. \end{aligned}$$

8.10 Punchline

Suppose M is an R-module. If $M_P = 0$, for every maximal ideal P, then $M = 0$. Even more importantly, a sequence is exact if and only if it is exact after localization at every maximal ideal.

So we can prove some global algebraic fact by looking at each maximal ideal separately, almost like proving continuity for a function.

8.11 Some problems

Problem 8.1. Consider

$$\mathbb{Z}_{(3)} = \left\{ \frac{a}{b} \;\middle|\; a, b \in \mathbb{Z}, 3 \nmid b \right\}.$$

Recall that this means we invert every integer outside (3).
Answer the following:

1. Decide which of the following are elements of $\mathbb{Z}_{(3)}$:

$$\frac{1}{2},\ \frac{1}{3},\ \frac{6}{7},\ \frac{9}{15}.$$

2. Which of the following are units in $\mathbb{Z}_{(3)}$?

$$2,\ 3,\ 6,\ \frac{3}{7},\ \frac{10}{21}.$$

3. Show that

$$\frac{6}{3} = \frac{2}{1}$$

is not a legitimate equality to use inside $\mathbb{Z}_{(3)}$ by simply treting $6/3$ as an element there. Explain why.

4. Describe the unique maximal ideal of $\mathbb{Z}_{(3)}$.

Solution. $1/2$ is an element of $\mathbb{Z}_{(3)}$ as 2 is not divisible by 3.

The same is true for $6/7$, while $1/3$ is not in $\mathbb{Z}_{(3)}$. $9/15 = 3/5$ so is in $\mathbb{Z}_{(3)}$.

2 is a unit since $1/2 \in \mathbb{Z}_{(3)}$ is the multiplicative inverse.

The same does not hold for the rest.

$3,\ 6$ are divisible by 3, $3/7$ has no inverse. The last element is not an element in $\mathbb{Z}_{(3)}$.

I believe $6/3$ is not a valid representative in the construction.

The unique maximal ideal is $(3)\mathbb{Z}_{(3)}$ which is

$$\left\{ \frac{a}{b} \mid 3 \mid a,\ 3 \nmid b \right\},$$

the set of nonunits. ■

Problem 8.2. Let $R = k[x]$ and $P = (x)$ where k is a field. Consider the localization

$$R_P = k[x]_{(x)}.$$

Since $P = (x)$, the denominators are the polynomials not divisible by x.
Work out the following:

1. Decide which of these are elements of R_P:

$$\frac{1}{1+x}, \frac{1}{x}, \frac{x^2+1}{x^2+x+1}, \frac{x}{x+x^2}.$$

2. Decide which of the following elements are units in R_P:

$$x,\ 1+x,\ x(1+x),\ \frac{x}{1+x},\ \frac{1+x}{1+x^2}.$$

3. Describe the unique maximal ideal of R_P.

Solution.

1.

$1/x+1$ is an element of R_P because $x+1 \notin (x)$ since otherwise we would have $x+1-x = 1 \in (x)$ which would imply that $(x) = k[x]$.

$$\frac{1}{x}$$

is not in R_P since $x \in (x)$, clearly.

We have $x^2, x \in (x)$ so, for the same reason as earlier, $x^2+x+1 \notin (x)$, meaning

$$\frac{x^2+1}{x^2+x+1} \in R_P.$$

Lastly, $x+x^2 \in (x)$ so $x/(x+x^2) \notin R_P$.

2.

x is not a unit since it's inverse would be $1/x$.

Meanwhile, $x+1$ has the inverse

$$\frac{1}{x+1}.$$

$x(1+x)$ is not a unit since its inverse would have x^2+x in its denominator. We know that no representative can satisfy this.

The fourth is not a unit, and the fifth is.

3.

The unique maximal ideal is $PR_P = (x)k[x]_{(x)}$. ■

Problem 8.3. Let $S = \{1, 6, 6^2, 6^3, \ldots\}$. Localize $\mathbb{Z}$ by

$$R_6 = S^{-1}\mathbb{Z} = \left\{ \frac{a}{6^n} \;\middle|\; a \in \mathbb{Z},\ n \geq 0 \right\}.$$

Work out the following:

1. Which of the following are units in R^6?

$$2,\ 3,\ 5,\ 6,\ 10,\ \frac{5}{6}.$$

2. For each prime ideal

$$(2),\ (3),\ (5),\ (7)$$

decide whetehr it is a prime ideal after localization.

Solution.

1.

In general, we use that $r/s \in R^6$ has an inverse if r is in the saturation of S, meaning there is some $q \in \mathbb{Z}$ such that $qr \in S$.

2 is a unit with inverse

$$\frac{3}{6}.$$

3 is a unit with inverse

$$\frac{2}{6}.$$

5 is not a unit as there does not exist any integer q such that $q \cdot 5 = 6^n$ for some n since 5 is not a factor of 6.

6 is a unit with inverse

$$\frac{1}{6}.$$

If 10 had an inverse then, since

$$10 = 2 \cdot 5$$

and $2^{-1} \in R_6$, we would require, by

$$(ab)^{-1} = b^{-1}a^{-1},$$

that 5 has an inverse, which it does not. Therefore 10 is not a unit.

We can also see that

$$\frac{5}{6}$$

has no inverse since 5 is not in the saturation of S.

2.

$2 \mid 6$ so $(2) \cap S \neq \emptyset$ and is therefore not prime in the localization.

Similarly, $3 \mid 6$ so (3) is not prime in the localization.

$5 \nmid 6^n$ for any $n \in \mathbb{N}$ so $(5) \cap S = \emptyset$ and is therefore prime.

Any 6^n is has a unique prime factor decomposition $(2 \cdot 3)^n$ so 7 cannot divide any power of 6 and thus $(7) \cap S = \emptyset$, meaning (7) must be prime. ■

Problem 8.4 (Set 6, Problem 4)**.** Let $A = k[x, y]$ be the polynomial ring in two variables, where k is a field. Let $I = (x, y)$ be the ideal generated by x and y.

a) Prove that $\phi : A^2 \to I$ defined by $(a, b) \mapsto ax + by$ is surjective and has kernel the submodule generated by the element $(-y, x)$.

b) Show that there is an exact sequence

$$0 \leftarrow I \xleftarrow{[x\ y]} A^2 \xleftarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} A \leftarrow 0$$

(Note that we then think of elements of A^2 as column vectors.)

Proof of (a). For $f \in I$, $f = p(x, y)x + q(x, y)y$.

Then, $p(x, y), q(x, y) \in A = k[x, y]$ so $(p(x, y), q(x, y)) \in A^2$.

Thus, for any $f \in I$ we can find $(a, b) \in A^2$ such that $\phi(a, b) = f$, meaning ϕ is surjective.

The submodule generated by $(-y, x)$ is

$$\{P(x, y)(-y, x) = (-P \cdot y, P \cdot x) \mid P \in A^2\}.$$

Then, for $(-P \cdot y, P \cdot x)$,

$$\phi(-P \cdot y, P \cdot x) = -Pyx + Pxy = 0.$$

Thus $(-y, x) \subseteq \ker \phi$.

Now suppose $(f, g) \in \ker \phi$. Then

$$0 = \phi(f, g) = fx + gy.$$

Then, since x and y are linearly independent, we must have that $f = \varphi y$ for some $\varphi \in k[x, y]$ and $g = \gamma x$ for some $\gamma \in k[x, y]$.

So

$$\varphi yx + \gamma xy = (\varphi + \gamma)xy = 0.$$

Since k is a field, we must have

$$\varphi + \gamma = 0$$

so

$$\varphi = -\gamma.$$

So (f, g) is necessarily of the form

$$(-Py, Px)$$

or

$$(Py, -Px)$$

for some $P \in k[x, y]$. Either way,

$$(f, g) \in (-y, x)$$

so

$$\ker \phi \subseteq (-y, x).$$

■

Problem 8.5 (Set 6, Problem 3)**.** Let k be a vector space, and let

$$0 \to k^n \to V \to k^m \to 0,$$

be a short exact sequence. Show that V is a vector space of dimension $n + m$.

Proof. Make the sequence more explicit:

$$0 \to k^n \xrightarrow{\iota} V \xrightarrow{\pi} k^m \to 0.$$

Since the sequence is exact, ι is injective, π is surjective, and $\ker \pi = \operatorname{im} \iota$.

Choose the standard basis $e_1, \dots, e_n$ of k^n. Since ι is injective, the vectors $u_i := \iota(e_i)$ are linearly indepdenent in V.

Next choose a standard basis $f_1, \dots, f_m$ of k^m.

Since π is surjective, we can choose some $v_j \in V$ such that $\pi(v_j) = f_i$.

We claim that the $n + m$ vectors

$$u_1, \dots, u_n, f_1, \dots, f_m$$

is a basis of V.

To show they generate V, let $v \in V$. Since $f_1, \dots, f_m$ is a basis of k^m,

$$\pi(v) = a_1 f_1 + \cdots + a_m f_m$$

for some $a_1, \dots, a_m$ in the ground field.

Therefore

$$\pi\left(v - \sum_{j=1}^{m} a_j v_j\right) = \pi(v) - \sum_{j=1}^{m} a_j \pi(v_j) = 0.$$

Hence

$$v - \sum_{j=1}^{m} a_j v_k \in \ker \pi.$$

By exactness, $\ker \pi = \operatorname{im} \iota$, so there is some $x \in k^n$ such that

$$v - \sum_{j=1}^{m} a_j v_j = \iota(x).$$

Writing

$$x = b_1 e_1 + \cdots + b_n e_n$$

we get

$$v = \sum_{i=1}^{n} b_i u_i + \sum_{j=1}^{m} a_j v_j.$$

For linear independence suppose

$$\sum_{i=1}^{n} b_i u_i + \sum_{j=1}^{m} a_j f_j.$$

Because $f_1, \ldots, f_m$ are all linearly independent,

$$a_1 = \cdots = a_m = 0,$$

Thus

$$\sum_{i}^{n} b_i e_i = 0,$$

but $u_i = \iota(e_i)$ are linearly independent since ι is injective. Hence

$$b_1 = \cdots = b_n = 0.$$

Thus these $n + m$ vectors form a basis, meaning $\dim_k V = n + m$. ■

9 Tying chapters 6 and 7 together

The main construction in chapter 6 is $R \rightsquigarrow S^{-1}R$, where $S \subseteq R$ is multiplicatively closed and you formally make every element of S invertible.

Algebraically, elements are fractions

$$\frac{a}{s},\ a \in R,\ s \in S,$$

with

$$\frac{a}{s} = \frac{a'}{s'}$$

when some $u \in S$ satisfies $u(as' - a's) = 0$.

The extra u is required for rings with zero divisors.

The important special cases are

$$R_a = S^{-1}R,\ S = \{1, a, a^2, a^3, \ldots\},$$

and especially

$$R_P,\ S = R \setminus P$$

for prime ideal P. In R_P, everything outside P becomes a unit. Gathmann emphasizes these as standard forms of localization.

The first major theorem to understand is that

$$\operatorname{Spec}(S^{-1}R) \Leftrightarrow \{P \in \operatorname{Spec} R \mid P \cap S = \emptyset\}.$$

So localization removes the primes that meet S.

For localization at a prime P, the surviving primes are precisely the primes contained in P. Hence R_P has a unique maximal ideal

$$PR_P,$$

so R_P is a local ring.

Then modules enter

$$S^{-1}M = \left\{ \frac{m}{s} \,\middle|\, m \in M,\ s \in S \right\}.$$

This connects directly back to tensors

$$S^{-1}M \cong M \otimes_R S^{-1}R.$$

So localization is extension of scalars from R to $S^{-1}R$.

Then exact sequences return,

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

remains exact after localization,

$$0 \to S^{-1}M_1 \to S^{-1}M_2 \to S^{-1}M_3 \to 0.$$

The payoff is the notion of a local property.

For example,

$$M = 0 \Leftrightarrow M_P = 0$$

for all maximal P, and exactness can also be checked after localizing at every maximal ideal.

9.1 Chapter 7

Here we introduce the notion of avoiding infinite complexity.

The two definitions are

Definition 9.1 (Notherian and Artinian)**.**

$$M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$$

must eventually stop for a **Notherian** module, while

$$M_0 \supseteq M_1 \supseteq M_2 \supseteq \cdots$$

must eventually stop for an **Artinian** module.

Crucially,

$$M \text{ is Notherian} \Leftrightarrow \text{every submodule of } M \text{ is finitely generated,}$$

a more useful characterization. Gathmann notes that there is no equivalent notion for Artinian modules.

Some useful examples to know are the following.

$\mathbb{Z}$ is Noetherian, but not Artinian since

$$\mathbb{Z} \supsetneq 2\mathbb{Z} \supsetneq 4\mathbb{Z} \supsetneq \cdots.$$

Meanwhile, an infinite polynomial ring $k[x_0, x_1, ...]$ is not Notherian because

$$(x_0) \subsetneq (x_0, x_1) \subsetneq (x_0, x_1, x_2) \subsetneq \cdots.$$

Then exact sequences return once more. If

$$0 \to N \to M \to M/N \to 0$$

is exact, then M is Noetherian if and only if N and M/N are Noetherian, and similarly for Artinian.

A useful fact is that

$$M \text{ has finite length} \Leftrightarrow M \text{ is both Noetherian and Artinian.}$$

Then we get to one of the most useful theorems.

Theorem 9.1. *Hilbert's basis theorem*

$$R \text{ Noetherian} \Rightarrow R[x] \text{ Noetherian}.$$

From this we can deduce that $R[x_1, \dots, x_n]$ is Noetherian whenever R is, and every finitely generated algebra over a Noetherian ring is Noetherian. In particular, coordinate rings of affine varieties are Noetherian.

This is a foundational result explaining, in part, why finite polynomial descriptions are so well-behaved.

The Artinian half is stronger and more rigid. For an Artinian ring,

- there are only finitely many prime ideals,
- every prime ideal is maximal,
- some product of maximal ideals is zero.

Then another useful theorem.

Theorem 9.2 (Hopkins' theorem)**.**

$$R \text{ Artinian} \Leftrightarrow R \text{ Noetherian and every prime ideal is maximal}.$$

Finally, every Artinian ring decompose as a finite product of Artinian rings

$$R \cong \prod_{i=1}^{n} R_{P_i},$$

where P_i are its maximal ideals.

9.2 Some problems

Problem 9.1. Let $R = k[x]$ where k is a field.
Prove that R is Noetherian, but not Artinian.

Proof. Since k is a field, it has only two ideals 0 and k.

Thus k is Noetherian, since a chain of ideals terminates finitely,

$$0 \subsetneq k = k.$$

Since k is Noetherian, $k[x] = R$ is Noetherian.

It is not Artinian since

$$(x) \supsetneq (x^2) \supsetneq (x^4) \supsetneq \cdots$$

does not terminate finitely. ■

Problem 9.2. Let R be a Noetherian ring and let $S \subseteq R$ be multiplicatively closed. Prove that

$$S^{-1}R \text{ is Noetherian.}$$

Proof. We will use the characterization that R is Noetherian if and only if every ideal of R is finitely generated.

Let J be some ideal of $S^{-1}R$.

Then

$$J^c = (r \in R \mid r/1 \in J)$$

is an ideal of R. Since R is Noetherian,

$$J^c = (a_1, \dots, a_n)$$

is finitely generated.

We claim that

$$J = (a_1/1, \dots, a_n/1).$$

First, since $a_i \in J^c$, by definition

$$\frac{a_i}{1} \in J.$$

Hence

$$(a_1/1, \dots, a_n/1) \subseteq J.$$

For the other inclusion, take

$$\frac{r}{s} \in J.$$

Because J is an ideal and $s/1 \in S^{-1}R$,

$$\frac{r}{1} = \frac{r}{1} \cdot \frac{r}{s} \in J.$$

Therefore $r \in J^c$.

Since $J^c = (a_1, \dots, a_n)$, there are $c_1, \dots, c_n \in R$ such that

$$r = c_1 a_1 + \cdots + c_n a_n.$$

Therefore,

$$\frac{r}{s} = \frac{c_1 a_1 + \cdots + c_n a_n}{s} = \sum_{i=1}^{n} \frac{c_i}{s} \frac{a_i}{1}.$$

Thus

$$\frac{r}{s} \in \left(\frac{a_1}{1}, \dots, \frac{a_n}{1}\right).$$

So

$$J = \left(\frac{a_1}{1}, \dots, \frac{a_n}{1}\right).$$

Since J was arbitrary, $S^{-1}R$ is Noetherian. ■

Problem 9.3. Let $R = k[x]/(x^4)$, where k is a field.
Prove that R is both Noetherian and Artinian.

Proof. Suppose J is an ideal of R. Then J is an ideal of $k[x]$ satisfying

$$(x^4) \subseteq J.$$

For the Noetherian property, we make the observation that $k[x]$ is a principal ideal domain so all ideals are principal and thus finitely generated.

Then, ideals J of R have $(x^4) \subseteq J$, as mentioned so

$$f \mid x^4.$$

Up to multiplication by a unit, the only divisors are $1, x, x^2, x^3, x^4$ so the ideals of R are exactly

$$R,\ (\overline{x}),\ (\overline{x}^2),\ (\overline{x}^3),\ (0).$$

Thus no decsending chain can be infinite since

$$R, \supsetneq (\overline{x}), \supsetneq (\overline{x}^2), \supsetneq (\overline{x}^3), \supsetneq (0).$$

So R is Artinian and, sicne all ideals are finitely generated, Noetherian.

The only prime ideal is $(\overline{x})$. ■

Problem 9.4. Let $R = \mathbb{Z}/12\mathbb{Z}$.

1. Determine all ideals of R.
2. Use the list to prove R is Noetherian and Artinian.
3. Determine all prime ideals of R, and check that each prime ideal is maximal.
4. Pick one of the prime ideals P, localize R at P, and try to identify R_P with a familiar quotient of $\mathbb{Z}$.

Proof. The ideals are those generated by divisors of 12, namely

$$0,\ R,\ (\overline{2}),\ (\overline{3}),\ (\overline{4}),\ (\overline{6}).$$

Then there are no infinite ascending or descending chains of ideals, clearly.

Therefore R is Noetherian and Artinian.

The prime ideals of R are those generated by the prime factors of 12,

$$(2) \text{ and } (3).$$

These are indeed maximal. Observe that

$$R/(\overline{2}) \cong \mathbb{Z}/2\mathbb{Z}$$

and

$$R/(\overline{3}) \cong \mathbb{Z}/3\mathbb{Z},$$

both of which are fields.

Let us localize at $(\overline{2})$.

We make $S = R \setminus (\overline{2})$ making

$$R_P = S^{-1}R = \left\{ \frac{a}{b} \;\middle|\; a \in R,\ 2 \nmid b \right\}.$$

Here we have the integers modulo 12 where all odd $\overline{x}$ are invertible.

In particular, $\overline{3} \notin P$ so $\overline{3}$ becomes a unit in R_P, but $\overline{3} \cdot \overline{4} = 0$ already.

Then

$$\overline{3}^{-1} \cdot \overline{3} \cdot \overline{4} = 0$$

so

$$\overline{4} = 0.$$

Thus

$$R_P \cong \mathbb{Z}/4\mathbb{Z}.$$

■

Problem 9.5. Let $R = k[x, y]/(xy)$, and write $\overline{x}, \overline{y}$ for the residue classes of x and y. Let

$$P = (\overline{x}) \triangleleft R.$$

Work through the following.

1. Show that P is a prime ideal.
2. Consider the localization R_P. Explain why $\overline{y}/1$ is a unit in R_P.
3. Use the relation $\overline{xy} = 0$ to prove that

$$\overline{x}/1 = 0.$$

4. Use this to identify R_P with a familiar field.
5. Finally, explain why R is Noetherian, and hence why R_P is Noetherian as well.

Proof. First we note that the residue class $\overline{x}$ is

$$x + (xy).$$

So

$$(\overline{x}) = \{\overline{xf} \mid f \in k[x, y]\}.$$

We will use the characterization that P is prime if and only if R/P is an integral domain.

By the third isomorphism theorem,

$$\frac{k[x, y]/(xy)}{(\overline{x})} \cong \frac{k[x, y]}{(xy, x)},$$

but since $xy \in (x)$ we have

$$(xy, x) = (x).$$

Therefore

$$R/(\overline{x}) \cong k[x, y]/(x) \cong k[y].$$

$k[y]$ is an integral domain since k is a field. Hence $(\overline{x})$ is prime.

Set $S = R \setminus (\overline{x})$. We consider the localization,

$$R_P = S^{-1}R = \left\{\frac{a}{b} \;\middle|\; a \in R,\ b \notin (\overline{x})\right\}.$$

We have $1 \notin (\overline{x})$ since otherwise we would have $(\overline{x}) = R$. Thus $\overline{y}/1$ is in the localization, and since $\overline{y}$ is not divisible by $\overline{x}$ we have

$$\frac{1}{\overline{y}} \in R_P.$$

So $\overline{y}/1$ is a unit in R_P.

In R, we have that

$$\overline{xy} = 0,$$

so in the localization

$$0 = 0 \cdot \frac{1}{\overline{y}} = \frac{\overline{x}}{1} \cdot \frac{\overline{y}}{1} \cdot \frac{1}{\overline{y}} = \frac{\overline{x}}{1}.$$

Then

$$R_P \cong \frac{k[x, y]}{(x)} = k[y].$$

As k is Noetherian we have that $k[y] \cong R_P$ is Noetherian. We could also make this con-

clusion from observing that $k[x, y]$ is Noetherian (for the same reason) and since quotienting preserves this the quotient R is, implying that R_P is Noetherian since localizing also preserves this property. ■

Problem 9.6. Let

$$R = k[x],\ M = R/(x^2) \oplus R/(x-1),$$

and let

$$P = (x) \lhd R.$$

Consider the localization M_P.
Show the following:

1. Using localization commuting with direct sums,
$$M_P \cong (R/(x^2))_P \oplus (R/(x-1))_P.$$
2. Prove that
$$(R/(x-1))_P = 0.$$
3. Show that
$$(R/(x^2))_P \cong R_P/(x^2)R_P.$$
4. Conclude that
$$M \cong R_P/(x^2)R_P.$$

Proof. In general, for a commutative ring R and some finite collection of R-modules $\{M_i\}_{i=1}^n$ we have that

$$\bigoplus_{i=1}^{n} S^{-1}M_i \cong S^{-1}\left(\bigoplus_{i=1}^{n} M_i\right).$$

As $R/(x^2)$ and $R/(x-1)$ are themselves rings, and thus modules, we have a canonical isomorphism

$$M = R/(x^2)_P \oplus R/(x-1)_P \cong (R/(x^2) \oplus R/(x-1))_P.$$

We have

$$(R/(x-1))_P = \left\{ \frac{f(x) + p(x)(x-1)}{q(x)} \;\middle|\; x \nmid q(x) \right\}.$$

Since $x - 1 = 0$ in $R/(x-1)$ we have $x \equiv 1$, meaning every element is divisible by x, so the localization is trivial.

In $(R/(x^2))_P$ we have that we make polynomials not divisible by x invertible, meaning constant polynomials (which are already invertible), as well as polynomials of degree ≥ 2.

For 3. consider the exact sequence

$$0 \to (x^2) \xrightarrow{\iota} R \xrightarrow{\pi} R/(x^2) \to 0$$

where ι is the inclusion map and π is the projection into the quotient.

Then, since localization preserves exactness,

$$0 \to (x^2)_P \to R_P \to (R/(x^2))_P \to 0$$

is exact. Then, since the image of the homomorphism from $(x^2)_P$ into R_P (the inclusion) is $(x^2)_P$ itself,

$$\frac{R_P}{(x^2)_P} \cong (R/x^2)_P.$$

Since $(R/(x-1))_P$ was trivial, the projection homomorphism

$$(R/(x^2))_P \oplus (R/(x-1))_P \to (R/(x^2))_P$$

is an isomorphism, hence

$$M \cong (R/(x^2))_P \cong R_P/(x^2)R_P.$$

■

Problem 9.7. Let $A = k[x,y]/(x^2-1, y^2-1)$.

a) Is A an Artinian ring?
b) What are the maximal ideals of A?
c) Write A as a direct product of artinian local rings.

Solution to a. AS $x^2 = 1$ and $y^2 = 1$, we have that polynomials in A have maximum degree 1 in both x and y, meaning

$$f(x,y) = a + bx + cy + dxy,\ a,b,c,d \in k.$$

We can observe that this can be interpreted as a finite dimensional k-algebra spanned by the images of the polynomials $\{1, x, y, xy\}$, meaning it is necessarily Artinian.

Since the ring is Artinian the maximal ideals correspond to the prime ideals. ■

Problem 9.8. Let

$$R = k[x,y]/(x^2, xy, y^3),$$

where k is a field. Write $\overline{x}, \overline{y}$ for the classes of x, y in R.
Prove the following,

1. Every element of R can be written uniquely in the form
$$a + b\overline{x} + c\overline{y} + d\overline{y}^2,\ a, b, c, d \in k.$$
2. Show that $\mathfrak{m} = (\overline{x}, \overline{y})$ is the unique maximal ideal of R.
3. Compute $\mathfrak{m}^2, \mathfrak{m}^3$.
4. Use this to prove that R is Artinian.
5. Find a composition series for R as an R-module and determine it's length
$$\ell_R(R).$$

Proof. Suppose
$$f \in R.$$
Then
$$f = q + (x^2, xy, y^3).$$
For $q \in k[x, y]$ we have that any terms involving both x and y will be in (x^2, xy, y^3).

For y degree ≥ 3 is zero and for x degree ≥ 2 is zero. The same holds for terms including only x or y.

Thus any $f \in R$ can be written in the form described.

We begin by proving that it is maximal by observing that
$$\frac{R}{(\overline{x}, \overline{y})}$$
is a field.

For $f \in R$ we have
$$f = a + b\overline{x} + c\overline{y} + d\overline{y}^2$$
so since all nonconstant terms are in $(\overline{x}, \overline{y})$ we intuitvely see that this should be isomorphic to k.

We have that, as ideals of $k[x, y]$,
$$(x^2, xy, y^3) \subseteq (x, y).$$
Furthermore,
$$\frac{(x, y)}{(x^2, xy, y^3)} = (\overline{x}, \overline{y})$$
so we can use the third isomorphism theorem,
$$\frac{R}{(\overline{x}, \overline{y})} = \frac{k[x, y]/(x^2, xy, y^3)}{(x, y)/(x^2, xy, y^3)} \cong \frac{k[x, y]}{(x, y)} \cong k.$$
Thus $(\overline{x}, \overline{y})$ is maximal. For uniqueness, suppose another ideal I of R is maximal.

For uniqueness we will show that every element in $R \setminus (\overline{x}, \overline{y})$ is a unit.

Take

$$f = a + b\overline{x} + c\overline{y} + d\overline{y}^2 \in R \setminus (\overline{x}, \overline{y}).$$

Then, since

$$\overline{x},\ \overline{y},\ \overline{y}^2$$

are all multiples of either $\overline{x}$ or $\overline{y}$, we get that they are in the maximal ideal and thus $\overline{0}$, so

$$f = a \in k$$

which is necessarily invertible since k is a field.

An element of $(\overline{x}, \overline{y})$ is of the form

$$a\overline{x} + b\overline{y}.$$

Then $\mathfrak{m}^2$ is the ideal of all 2-fold products

$$\begin{aligned}(a\overline{x} + b\overline{y})(c\overline{x} + d\overline{y}) &= ac\overline{x}^2 + ad\overline{xy} + bc\overline{xy} + bd\overline{y}^2 \\ &= 0 + 0 + 0 + bd\overline{y}^2\end{aligned}$$

so we simply have that

$$\mathfrak{m}^2 = (\overline{y}^2).$$

For $\mathfrak{m}^3$ this is similar

$$\begin{aligned}a\overline{y}^2(b\overline{x} + c\overline{y}) &= ab\overline{y}^2\overline{x} + ac\overline{y}^3 \\ &= 0.\end{aligned}$$

Thus, as an a module over itself, we get a chain

$$0 = \mathfrak{m}^3 \subsetneq \mathfrak{m}^2 \subsetneq \mathfrak{m} \subsetneq R$$

making R have finite length, implying its Artinian. ■

Problem 9.9. Let R be a Noetherian ring, and let M be a finitely generated R-module.
Prove that

$$M \text{ is Noetherian.}$$

Proof. Choose generators $m_1, \dots, m_n$ for M and define

$$\varphi : R^n \to M,\ (a_1, \dots, a_n) \mapsto \sum_{i=1}^{n} a_i m_i.$$

This map is surjective, since for any $m \in M$ we have

$$m = \sum_{i=1}^{n} a_i m_i$$

since M is finitely generated.

Thus we have $(a_i)_{i=1}^n \subseteq R^n$ mapping to m by φ.

We do not note $\ker \varphi = 0$ since,

$$\sum_{i=1}^{n} a_i m_i = 0$$

does not impliy $a_i = 0$ for all i, given that the module is not specified to be free.

We get a short exact sequence

$$0 \to \ker \varphi \to R^n \to M \to 0$$

by $(\ker \varphi \to R^n)(x) = x$.

Thus

$$\operatorname{im}(0 \to \ker \varphi) = 0 = \ker(\ker \varphi \to R^n).$$

Next

$$\operatorname{im}(\ker \varphi) = \ker \varphi.$$

Next

$$\operatorname{im} \varphi = M$$

since φ is surjecitve. Lastly,

$$\ker(M \to 0) = M.$$

So the sequence is exact.

We have that R is Noetherian. Arguing inductively we can suppose that R^k is Noetherian and construct a short exact sequence of R-modules

$$0 \to R \to R^{k+1} \to R^k \to 0$$

where R embeds into the first coordinate and the quotient

$$R^{k+1}/R \cong R^k.$$

Since submodules and quotient modules remain Noetherian, the middle module is Noetherian.

Thus we can use this to conclude that M is Noetherian. ■